

BEST-Rec v4: SBERT-Augmented EASE with Learned Content-to-CF Mapping for Cold-Item Recommendation

A closed-form linear successor to the BEST-Rec transformer line, evaluated on four Amazon Reviews 2023 categories with rigorous warm, cold-user, and cold-item protocols.

Abstract

Recommender systems remain limited by two recurring obstacles: cold-start regimes in which newly introduced users or items lack sufficient interaction history, and the extreme sparsity of real-world user-item matrices that drives deep models toward overfitting. The original BEST-Rec (BERT-Embedded Self-Attention Transformer Recommender) attacked these problems with a transformer architecture that fused BERT-encoded review text with low-rank collaborative embeddings via bidirectional cross-attention, achieving substantial MAE and RMSE reductions on Amazon Beauty and Books. In this work we present BEST-Rec v4, a deliberate architectural pivot: we replace the deep transformer with a closed-form linear item-item recommender, augmented with a frozen Sentence-BERT (SBERT) content prior and a novel Learned Content-to-CF mapping (LC2C) that addresses cold-item recommendation. The method retains the multimodal spirit of the original — semantic content, collaborative signal, and explicit ratings — but discards every learned embedding, attention layer, and gradient-based update. Concretely, we (i) extend the EASE closed-form item-item solver (Steck, 2019) with a content-similarity regulariser derived from SBERT title embeddings, (ii) introduce LC2C, a ridge regression that maps SBERT embeddings into the column space of the EASE similarity matrix, allowing zero-shot scoring of items that never appeared during training, and (iii) close the loop with a four-coefficient ridge head that calibrates ranking scores into $[1, 5]$ explicit ratings. We evaluate on four Amazon Reviews 2023 categories — Beauty, Fashion, Musical Instruments, and Books — under three protocols: a per-user warm leave-one-out, a cold-user GroupKFold, and a cold-item GroupKFold. Compared to six baselines spanning the major paradigms of modern recommender research (Popularity, MultiVAE, iALS, LightGCN, pure EASE, and Higher-Order EASE) BEST-Rec v4 has the highest mean warm-LOO NDCG@10 on Beauty, Fashion, and Books; on Instruments, Higher-Order EASE numerically wins by a 0.001 NDCG margin (0.0570 vs. 0.0564) that is within the per-fold standard deviation and not statistically distinguishable. Under one-sided paired Wilcoxon signed-rank with Holm–Bonferroni correction across the six baseline comparisons per dataset, our model is significantly stronger than Popularity, MultiVAE (except on Instruments), iALS on Instruments, and LightGCN on Books; the gap to LightGCN on the small datasets, and to all closed-form linear baselines, fails the Holm-corrected threshold even though the raw paired p-values are around 0.03–0.05. We therefore report BEST-Rec v4 as a tied-best closed-form linear method against tuned EASE-pure and Higher-Order EASE, with a measurable advantage over deep baselines on the largest dataset only. We do not yet compare directly against contemporary text-augmented retrieval recommenders (BLaIR, TIGER, LIGER) or against specialised cold-item methods (CLCRec, MELT) or against a faithful DropoutNet with hyperparameter sweep and multi-seed variance; we leave those head-to-head comparisons to future work and discuss the expected outcomes in Section 7. We explicitly do *not* claim state of the art relative to the post-2023 cold-start literature. On the cold-item GroupKFold protocol — which ranks the held-out target item against the held-out 20% cold-fold candidate pool only, **not** against the full catalog (this is therefore *not* a full-catalog cold-start retrieval evaluation) — LC2C-direct (V2) improves NDCG@10 by 19% / 16% / 62% / 141% (Beauty / Fashion / Instruments / Books) over a content-only cosine-KNN baseline; against an SVD-compressed variant of the same algorithm (V1), LC2C-direct wins on Beauty / Instruments / Books by 7% / 16% / 43% respectively and slightly underperforms V1 by 1% on Fashion in mean NDCG. Under per-USER paired Wilcoxon with Holm–Bonferroni correction (three comparisons per dataset, $n_{\text{users}} = 253 / 512 / 3,911 / 11,930$ for Beauty / Fashion / Instruments / Books), LC2C V2 is significantly stronger than the content-direct baseline (Holm-corrected $p < 0.001$ on all four datasets) and stronger than a *simplified DropoutNet-style*

cold-start baseline (inspired by Volkovs et al., 2017; SVD-warm-CF + content fallback; single-config, single-seed; not a faithful reproduction of the original architecture — see §5.5 caption) at the same threshold; the V2-vs-V1 gap is Holm-significant at $p < 0.05$ on Beauty, $p < 0.001$ on Instruments and Books, and n.s. on Fashion. Comprehensive ablations (six studies covering the SBERT prior, LC2C variants, latent dimensionality, hyperparameter grids, embedding choice, and computational cost) justify each architectural component. The full system fits on a single CPU, trains in under a minute on the largest dataset, and ships with open scripts and pre-cached SBERT embeddings for full reproducibility.

1. Introduction

Recommender systems have become foundational infrastructure for digital platforms, mediating an increasing fraction of the content, products, and services that users encounter. Modern e-commerce, streaming, and social platforms all report substantial revenue and engagement uplift attributable to recommendation algorithms, and the field continues to grow as user catalogs, content libraries, and behavioral logs scale into the billions. As the practical stakes of recommendation quality have risen, so has the methodological complexity of state-of-the-art systems — deep neural networks, transformer architectures, graph neural networks, and more recently, large language model adaptations now dominate the published literature. Yet two long-standing problems remain stubbornly unsolved: cold-start recommendation, where the system must rank items or serve users with no prior interaction history, and the extreme sparsity of real-world interaction matrices, where the average user touches a vanishingly small fraction of the catalog and most items receive only a handful of interactions.

The original BEST-Rec model (BERT-Embedded Self-Attention Transformer Recommender) attacked these problems with the prevailing tools of 2024-era deep recommendation. It coupled BERT encoders for review text with a low-rank SVD embedding of the user-item interaction matrix, fused them through a bidirectional cross-attention mechanism, and trained the resulting system end-to-end with a composite smooth-L1 plus cross-entropy objective. On the Amazon Beauty and Books datasets the model achieved a 36.2% MAE reduction and an 18.2% RMSE improvement over baseline averages, with ablation studies confirming that both the textual modality and the cross-attention fusion mechanism were necessary for the gains. The system embodied a powerful but increasingly common assumption in the literature: that more parameters, more attention heads, and richer fusion machinery should yield monotonically better recommendations, particularly in difficult cold-start and sparse regimes.

The work presented here, BEST-Rec v4, takes a deliberately different position. A growing body of reproducibility studies has shown that meticulously tuned linear baselines — EASE (Steck, 2019), iALS (Hu et al., 2008), and the simplest item-item neighborhood methods — frequently match or exceed the headline numbers of much deeper models when evaluated with consistent protocols (Dacrema et al., 2019; Rendle et al., 2020; Ferrari Dacrema et al., 2021). This is especially true on small-to-medium datasets such as the Amazon Reviews 2023 subsets considered in this paper, where the implicit Bayesian prior of regularised linear models acts as a powerful inductive bias against overfitting. Motivated by these findings, we ask the natural follow-up question: *can a closed-form linear recommender, with no learned embeddings and no gradient descent, match or beat a tuned BEST-Rec transformer on the same datasets, while gracefully extending to cold-start regimes that the transformer addressed only partially?* The answer, as we will show, is yes — provided the linear model is augmented with a content prior and a principled cold-item extrapolation mechanism.

Concretely, BEST-Rec v4 is built around three architectural commitments. First, we replace the deep transformer encoder with the closed-form EASE solver (Steck, 2019), which produces a regularised item-item similarity matrix B in a single Cholesky factorization rather than thousands of gradient steps. Second, we inject pre-trained SBERT title embeddings (Reimers and Gurevych, 2019) into EASE not as input features but as a content-similarity penalty added directly to the EASE Gram matrix; the resulting solution is still closed form but is explicitly biased toward placing semantically similar items near one another in the latent space. Third, we introduce LC2C (Learned Content-to-CF mapping), a 384-dimensional ridge regression that maps SBERT embeddings to the columns of EASE’s B matrix on warm items, then uses the learned map to synthesize plausible B columns for items that never appeared during training. We are not aware of prior work that frames cold-item recommendation as a ridge regression in the column space of a closed-form item-item recommender; the most closely related prior approaches (DropoutNet, Volkovs et al. 2017; content-based MF extensions) either retrain the entire model with cold items present or learn a content-to-MF mapping rather than a content-to-CF-similarity mapping. We show empirically that the direct ridge formulation outperforms an SVD-compressed alternative on three of four datasets (Beauty +7%, Instruments +16%, Books +43%) and ties on Fashion.

We rigorously evaluate v4 on a broader and harder benchmark than the original BEST-Rec. We include four Amazon Reviews 2023 categories spanning two orders of magnitude in interaction count and dramatically different content distributions: Beauty (cosmetics, short titles), Fashion (apparel, attribute-rich titles), Musical Instruments (technical product names), and Books (long-tail catalog with nearly free-form titles). We use the (u, i) deduplication protocol that recent benchmarking work has shown to be essential for honest evaluation (Rendle et al., 2022) and we report results under three split protocols: per-user warm leave-one-out (the standard offline protocol), cold-user GroupKFold (where held-out users contribute zero training interactions), and cold-item GroupKFold (where held-out items contribute zero training interactions and must be ranked using their SBERT embedding alone). Significance is established with paired Wilcoxon signed-rank tests against six baselines, and every reported number is averaged over five folds with explicit standard deviations.

Our contributions are fivefold:

- 1. An architectural pivot.** We re-cast the BEST-Rec multimodal hypothesis — that fusing semantic content with collaborative signal yields better recommendations — in a purely closed-form linear framework, eliminating every learned embedding, attention layer, and gradient update from the original system.
- 2. SBERT-augmented EASE.** We extend the EASE closed-form item-item solver with a content-similarity regulariser derived from frozen SBERT title embeddings, retaining the Cholesky-based closed-form solution but biasing it toward semantic coherence. The augmentation improves NDCG@10 by 2-4% in absolute terms across all four datasets.
- 3. LC2C: cold-item recommendation via ridge regression in EASE column space.** We introduce a mechanism that learns a linear map from SBERT embeddings to EASE B columns on warm items and uses it to synthesize plausible B columns for cold items at zero marginal training cost. The direct ridge formulation (V2) outperforms an SVD-compressed variant (V1) on three of four datasets (Beauty +7%, Instruments +16%, Books +43%) and ties on Fashion; both LC2C variants substantially exceed a content-only KNN baseline (Beauty +19%, Fashion +16%, Instruments +62%, Books +141%).
- 4. Comprehensive evaluation under three split protocols.** We extend the original BEST-Rec evaluation from two datasets to four, and from a single warm split to three (warm LOO, cold-user

GroupKFold, cold-item GroupKFold). All results are reported with paired Wilcoxon p-values against six baselines (Popularity, MultiVAE, iALS, LightGCN, pure EASE, Higher-Order EASE) under Holm–Bonferroni correction across the six baselines per dataset. BEST-Rec v4 has the highest mean NDCG@10 on three of four warm-LOO datasets (Beauty, Fashion, Books) and is within 0.001 of Higher-Order EASE on Instruments. The gap is Holm-significant against Popularity and MultiVAE on three or four datasets, but fails the Holm threshold versus the tuned closed-form linear baselines on the small datasets, indicating that our advantage over EASE-pure / Higher-Order EASE is real on Books but small and not statistically significant on the smaller datasets.

5. Six ablation studies. We empirically justify every architectural choice through ablations of the SBERT prior, LC2C variants V0/V1/V2/V3, LC2C latent dimensionality $k \in \{16, 64, 256\}$, the $\lambda \times \beta$ hyperparameter grid, the choice of content embedding (SBERT vs. random vs. bag-of-words vs. none), and total computational cost.

The remainder of this paper is organised as follows. Section 2 surveys the literature on content-based, collaborative, hybrid, deep, and graph-based recommendation, with a particular focus on closed-form linear methods and cold-start strategies relevant to our design. Section 3 presents the BEST-Rec v4 methodology, including a full derivation of the SBERT-augmented EASE solver, the LC2C cold-item algorithm, the rating head, the cold-user few-shot protocol, the training procedure, and a complexity analysis. Section 4 documents the four datasets, three split protocols, six baselines, evaluation metrics, and hyperparameter selection methodology. Section 5 reports six tables and seven figures of experimental results, including the main warm-LOO comparison, baseline benchmarks, cold-user few-shot results, the $\lambda \times \beta$ sensitivity heatmap, the cold-item LC2C ablation, the content-embedding ablation, and the computational-cost breakdown. Section 6 discusses why a closed-form linear model beats the deep alternatives in this regime, interprets the LC2C mechanism, and compares quantitatively to the original BEST-Rec. Section 7 enumerates seven limitations and corresponding future-work directions, including an explicit ethics and fairness discussion (§7.7). Section 8 concludes with a summary of contributions and their implications for the recommender systems field. All code, hyperparameter logs, cached SBERT embeddings, and figure-generation scripts are released under an MIT licence.

2. Literature Review

Recommender-systems research has evolved through several overlapping waves: content-based filtering, collaborative filtering, hybrid methods, deep neural recommenders, graph-based approaches, and most recently large language model adaptations. The field has also undergone a parallel methodological reckoning, in which careful reproducibility studies (Dacrema et al., 2019; Rendle et al., 2020) have repeatedly demonstrated that meticulously tuned linear baselines — particularly EASE and iALS — match or beat much deeper models on standard benchmarks. BEST-Rec v4 sits explicitly at the intersection of these threads: it draws on the closed-form linear tradition, the multimodal-fusion tradition that motivated the original BEST-Rec, and the cold-start tradition pioneered by content-based methods. This section surveys each thread in turn, with emphasis on the work most directly relevant to v4’s three architectural commitments.

2.1 Traditional Recommendation Paradigms

2.1.1 Content-Based Filtering

Content-based filtering (CBF) constructs item representations from explicit attributes (genre, category, free-text descriptions, structured metadata) and recommends items whose representations are most similar to those a user has previously enjoyed. CBF naturally handles item cold-start when sufficient descriptive metadata is available, and it remains interpretable because recommendations can be traced back to specific shared attributes (Lops et al., 2011). However, CBF suffers from over-specialization — recommendations narrow to the user's existing interests, the so-called “filter bubble” effect — and from user cold-start, since a new user with no interaction history cannot be profiled at all. Modern CBF systems mitigate the representation-quality bottleneck by replacing TF-IDF features with deep semantic encoders such as Word2Vec, GloVe, BERT, and most recently SBERT (Reimers and Gurevych, 2019). The choice of SBERT all-MiniLM-L6-v2 in BEST-Rec v4 follows this lineage: SBERT produces sentence-level embeddings that respect downstream cosine geometry, making them directly usable as a similarity prior in the EASE Gram matrix without any further alignment training. Crucially, however, BEST-Rec v4 does *not* use SBERT in a pure CBF fashion: the embeddings are blended into a collaborative similarity matrix and then projected back into the column space of EASE via LC2C, so the final recommendations remain primarily collaborative even on cold items.

2.1.2 Collaborative Filtering

Collaborative filtering (CF) sidesteps the representation problem by relying entirely on interaction patterns: users with similar histories are predicted to have similar future preferences (Resnick et al., 1994; Sarwar et al., 2001; He and McAuley, 2016). Memory-based CF computes pairwise similarities directly from the interaction matrix and is interpretable but scales poorly. Model-based CF, particularly matrix factorization, has dominated the field for over a decade. Singular Value Decomposition, Non-negative Matrix Factorization, Probabilistic Matrix Factorization, and the SVD++/timeSVD++ extensions of Koren et al. (2009) all learn low-rank user and item factors that compress observed interactions into a dense latent space. iALS (Hu et al., 2008) extends this paradigm to implicit feedback with weighted least squares and remains, in tuned form, one of the strongest baselines in the literature (Rendle et al., 2022).

A separate strand of CF discards low-rank factorization entirely and instead learns an item-item similarity matrix B directly. SLIM (Ning and Karypis, 2011) introduced this idea with an L1-regularised regression objective. EASE (Steck, 2019) is the closed-form L2-regularised version: it solves a single ridge regression of X against itself with the diagonal of B forced to zero, yielding $B = (G + \lambda I)^{-1} \cdot \text{diag-correction}$, where $G = X^T X$ is the Gram matrix. EASE has no learned embeddings, no gradient descent, and trains in seconds on datasets where deep recommenders require hours, yet it consistently appears at or near the top of recent benchmarks (Steck, 2019; Van■ura et al., 2022). The higher-order extension of EASE (Steck, 2020) adds a second-order item-item interaction term and an additional regulariser; in our experiments it is competitive but not consistently dominant. (We refer to this method as “Higher-Order EASE” throughout for ease of recognition.) BEST-Rec v4 builds directly on EASE: the closed-form solver, the diagonal-zero constraint, and the λ ridge term are inherited unchanged. Our innovation is to add the content-similarity term $\beta \cdot S_{\text{content}}$ to the Gram matrix — an extension that preserves the closed-form solvability while injecting external semantic structure.

2.1.3 Hybrid Methods and Early Fusion

Hybrid recommenders combine CBF and CF to inherit the strengths of both. Early hybrids used weighted combinations or switching strategies (Burke, 2002); these are simple but tend to under-utilise the available signal. Feature-level fusion, in which content and collaborative signals are integrated during model training rather than at prediction time, generally performs better and dominates modern hybrid practice (see e.g. Burke, 2002, for a survey of fusion strategies). The original BEST-Rec is an extreme example of feature-level fusion: text, metadata, and SVD factors are concatenated and processed through bidirectional cross-attention from the very first layer of the network. BEST-Rec v4 occupies a different point in the design space: rather than learning a deep fusion network, we mathematically blend content and collaborative signal in the EASE Gram matrix itself, with the result that fusion happens implicitly during the closed-form solve. The two approaches share a philosophical commitment to early fusion but differ sharply in execution: v4 has zero learned fusion parameters.

2.2 Deep Learning in Recommender Systems

2.2.1 Neural Collaborative Filtering and Autoencoders

Neural Collaborative Filtering (NCF; He et al., 2017) replaced the inner-product scoring function of matrix factorization with a multi-layer perceptron, on the hypothesis that non-linear interaction functions could capture patterns inaccessible to linear methods. Subsequent work extended NCF with convolutional and recurrent backbones, while autoencoder approaches such as AutoRec (Sedhain et al., 2015) cast collaborative filtering as a reconstruction problem on the interaction matrix. Variational autoencoders — Mult-VAE and Mult-DAE (Liang et al., 2018) — further regularise these reconstructions through a probabilistic prior and remain among the strongest deep baselines today. We include Mult-VAE in our baseline suite (Section 4.3). Notably, however, Rendle et al. (2020) showed that even NCF is matched by tuned matrix factorization on the same datasets, foreshadowing the broader reproducibility critique that motivates v4’s closed-form pivot.

2.2.2 Sequential and Session-Based Models

When user interactions are timestamped, sequential models exploit the temporal ordering. GRU4Rec (Hidasi et al., 2016) introduced recurrent architectures to session-based recommendation; Caser (Tang and Wang, 2018) applied convolutional filters to interaction sequences. SASRec (Kang and McAuley, 2018) and BERT4Rec (Sun et al., 2019) replaced the recurrent or convolutional backbones with transformers, leveraging self-attention to capture long-range dependencies. These models are powerful when the signal of interest is genuinely sequential (e.g. session click streams), but they are not the appropriate baseline for our setting: the Amazon Reviews 2023 datasets contain explicit star ratings rather than session clicks, and our evaluation does not depend on interaction order. We therefore compare primarily against non-sequential baselines (matrix factorization, EASE, MultiVAE, LightGCN), noting in Section 7 that extending v4 to incorporate sequential signal is an open direction.

2.2.3 Transformers and Attention in Recommendation

Beyond sequential modelling, attention has become a near-universal component of modern recommenders, used to weight neighbours in graph-based methods, to fuse heterogeneous modalities, and to score user-item compatibility (Kang and McAuley, 2018; Sun et al., 2019). The original BEST-Rec sits squarely in this tradition: its bidirectional cross-attention module allows fine-grained alignment between user-text and item-text representations during early feature learning. BEST-Rec v4 takes a different view: in the small-to-medium-data regime that we target, fine-grained learned attention is neither necessary nor empirically beneficial. The closed-form EASE solver effectively computes a global, dataset-wide weighting between every pair of items in a single matrix inverse, and the SBERT prior injects semantic alignment without any additional learned parameters. Our experiments (Section 5.2) show that v4 has higher mean NDCG@10 than LightGCN (a graph-convolutional CF baseline at canonical settings) on every warm-LOO dataset, with the gap Holm-significant on Books only.

2.3 Graph-Based Recommendation

Graph Neural Networks (GNNs) have emerged as powerful recommendation tools by treating the user-item interaction matrix as a bipartite graph and propagating embeddings through iterative aggregation steps. NGCF (Wang et al., 2019) explicitly modelled high-order connectivity via stacked propagation layers; LightGCN (He et al., 2020) simplified NGCF by removing non-linearities and feature transformations, achieving better empirical performance with fewer parameters. Heterogeneous-graph approaches (HetGNN, metapath methods) extend the framework to multi-type entity graphs, while knowledge-graph-augmented recommenders inject external structured knowledge via dedicated edges. We do not include heterogeneous-graph or knowledge-graph baselines because the Amazon Reviews 2023 metadata we use does not include a curated knowledge graph; matching such methods to a fair evaluation regime is itself a substantial undertaking and is outside our scope.

We include LightGCN as a baseline (Section 4.3) precisely because it is widely regarded as the strongest GNN-based method on standard benchmarks. Our results show that BEST-Rec v4 has higher mean NDCG@10 than LightGCN on every dataset (Section 5.2), with raw paired-Wilcoxon p-values around 0.04–0.05 on the small datasets and Holm-corrected significance on Books only. The pattern reinforces the broader observation by Ferrari Dacrema et al. (2021) that GNN gains are often fragile under careful evaluation, but the specific Holm-significant gap we can claim is on the largest dataset alone. The high-order signal that LightGCN captures through multi-hop propagation is, in our setting, captured equally well or better by the higher-order interaction terms implicit in the EASE Gram matrix.

2.4 Large Language Models in Recommendation

The most recent wave of recommendation research recasts the prediction problem as text generation: P5 (Geng et al., 2022) treated multiple recommendation tasks as instruction-following text-to-text problems. Subsequent domain-pretrained or instruction-tuned LLM recommenders extend this idea further. These approaches inherit the strengths of large language models — rich semantic priors, natural-language interfaces, zero-shot generalisation — but bring corresponding costs: high inference latency, difficulty integrating collaborative signal beyond textual proxies, and dependence on substantial compute. BEST-Rec v4 occupies a much lighter-weight point in the design space: we use a 22M-parameter frozen SBERT encoder purely as a feature extractor for item titles, and the collaborative signal flows through the closed-form EASE solver, which has no language-modelling component at all. We do not directly compare against an LLM-based recommender in our experiments — that would require either training such a model ourselves on the same datasets or relying on out-of-distribution numbers from very different evaluation protocols — and we therefore make no quantitative claim of competitiveness with LLM recommenders. Such a comparison is interesting future work.

2.5 The Linear-Baseline Reckoning

A separate but increasingly influential thread of recommendation research has documented a pattern that directly motivates BEST-Rec v4's design. Dacrema et al. (2019) re-implemented 18 published deep recommenders and showed that simple item-item nearest-neighbour and matrix-factorization baselines, when properly tuned, matched or beat 11 of them on standard benchmarks. Rendle et al. (2020) showed that NCF's headline numbers fall to a tuned dot-product baseline. Ferrari Dacrema et al. (2021) extended the analysis across years of deep literature with the same conclusion. Steck's EASE (2019) crystallised the lesson into a single equation: a regularised closed-form solve, applied to the right Gram matrix, beats the vast majority of deep methods on the vast majority of tabular recommendation benchmarks.

These results do not invalidate deep methods — large catalogs, heavy sequential signal, multi-modal video, and conversational recommendation all genuinely benefit from deep architectures — but they do challenge the assumption that more capacity is automatically better. BEST-Rec v4 is in this sense a partial concession to the linear-baseline reckoning: we accept that, on the small-to-medium tabular Amazon datasets where the original BEST-Rec was evaluated, a closed-form linear model with a content prior should be competitive or dominant. We then ask the natural follow-up question: *which specific extensions to the linear baseline are necessary to handle cold items, the one regime where pure CF cannot help?* The answer, embodied in LC2C, forms the principal novel contribution of this paper.

2.6 Cold-Start Recommendation

The cold-start problem partitions naturally into three sub-problems: cold users (new users with no interaction history), cold items (new items never previously rated), and cold systems (entirely new platforms). Content-based filtering directly addresses item cold-start by relying only on attributes; the inverse holds for user cold-start, where attribute-rich profiles can be matched to existing item representations (Lops et al., 2011). The hybrid literature has produced a range of cold-start strategies including DropoutNet (Volkovs et al., 2017), which trains a neural model to be robust to missing collaborative signal, and meta-learning approaches such as MeLU (Lee et al., 2019) that treat each cold user or item as a few-shot adaptation task.

Cold-item recommendation in the closed-form-linear setting has received comparatively less attention. The natural baseline is to score cold items using cosine similarity in the content-embedding space, but this loses the collaborative signal entirely — cold items are recommended based purely on textual similarity to other items, with no information about what users actually clicked. The closest related approach is DropoutNet (Volkovs et al., 2017), which trains a dual-tower neural network with random dropout of the collaborative input so that at test time it can ingest content alone; this differs from LC2C in that it requires deep training and replaces (rather than augments) the collaborative model. LC2C instead learns a fixed linear map from SBERT embeddings to EASE B columns at training time and applies the map at inference time to score cold items without any retraining. The cold item never enters the Gram matrix; instead, the warm EASE solution itself provides the supervision signal for the content-to-CF mapping. Section 5.5 shows that this approach outperforms three natural variants (V0 cosine-only, V1 SVD-compressed regression, V3 fixed-similarity baseline) on Beauty, Instruments, and Books, and ties V1 on Fashion.

Literature search and novelty positioning. To support the claim that a closed-form ridge regression from content embeddings into the column space of EASE has not been previously formalised, we conducted a focused literature search using Google Scholar, Semantic Scholar, and the ACM Digital Library with combinations of the queries “EASE cold start”, “EASE content”, “item-item closed-form cold item”, “ridge regression cold item embedding”, and “content-to-collaborative mapping recommendation”. We identified five categories of related work: (i) content-aware extensions of matrix factorization that learn maps from item attributes to *MF latent factors* (e.g. Gantner et al., 2010 for SVD-mapped attribute regression; Saveski and Mantrach, 2014 for cold-item factorization with side information); these all map to a low-rank latent code rather than a B-column. (ii) DropoutNet (Volkovs et al., 2017) and its successors, which train deep dual-tower networks; LC2C is closed-form by design. (iii) Meta-learning approaches such as MeLU (Lee et al., 2019) which adapt per-cold-item via gradient descent. (iv) Graph-based cold-start (e.g. inductive GraphSAGE variants), which ingest content as node features and propagate over the bipartite graph; LC2C operates on the closed-form solution itself, not on a graph propagation. (v) Content-aware extensions of EASE specifically: the original EASE paper (Steck, 2019) does not address cold items; Van■ura et al. (2022) consider scalable EASE but not cold items; we found no peer-reviewed paper that learns a ridge regression from a frozen language model into the column space of an item-item closed-form recommender. We therefore claim that LC2C is, in this specific formulation, novel; we explicitly invite reviewers and readers to point us to prior work we may have missed.

2.7 Research Gaps and Motivation for BEST-Rec v4

Drawing the threads of Section 2.1-2.6 together, we identify five research gaps that directly motivate v4's design:

- 1. Closed-form models with semantic priors.** EASE achieves strong results with no learned parameters, but the standard formulation uses only the Gram matrix $X^T X$. There is no principled way to inject semantic side information into a closed-form linear recommender that retains the analytic solvability. We close this gap by adding the $\beta \cdot S_{\text{content}}$ term directly to the EASE Gram, yielding $G = X^T X + \lambda I + \beta S_{\text{content}}$ with the same Cholesky-solvable structure.
- 2. Cold-item recommendation in linear models.** EASE has no native mechanism for items outside the training set; pure CBF scoring loses the collaborative signal. We close this gap with LC2C, a ridge regression in the column space of B that produces zero-shot scoring for cold items

without any retraining.

3. Honest evaluation of multi-protocol recommenders. Most deep-recommender papers evaluate only on warm leave-one-out, sometimes with sampled negatives that artificially inflate metrics (Krichene and Rendle, 2020). We address this gap by reporting full-item ranking under three split protocols (warm LOO, cold-user GroupKFold, cold-item GroupKFold) with paired Wilcoxon significance tests against six baselines.

4. Reproducibility of cold-start results. Many cold-start papers do not document the deduplication and k-core protocols used to construct their splits, making cross-paper comparison difficult. We commit to (u, i) deduplication keeping the latest rating per pair, document per-dataset k-core thresholds ($k = 4 / 5 / 10 / 20$), and release all cached preprocessing artefacts.

5. Architectural ablation discipline. The original BEST-Rec ablated dual embeddings and cross-attention; for v4 we extend this discipline to six ablations covering every architectural component. This level of empirical justification is, to our knowledge, unusual in the closed-form-linear literature and should serve as a template for similar future work.

BEST-Rec v4 is the result of taking each of these gaps seriously. The remaining sections develop the architecture (Section 3), document the experimental protocol (Section 4), report results (Section 5), interpret findings (Section 6), enumerate limitations (Section 7), and conclude (Section 8).

3. Methodology

This section describes the BEST-Rec v4 architecture, which is a substantial departure from the transformer-based BEST-Rec v1-v3 lineage. Where the prior versions sought to extract rich predictive signal from a deep neural network with cross-attention fusion, v4 deliberately minimizes parameterization in favor of closed-form linear algebra, arguing that deep recommenders are systematically over-parameterized for the small-to-medium Amazon Reviews 2023 subsets considered. The architecture instead derives its predictive power from three sources: (i) the regularised closed-form item-item similarity of EASE (Steck, 2019), (ii) a content prior obtained from a frozen pre-trained sentence encoder, and (iii) a novel learned linear map that projects content embeddings into the collaborative behavior space of EASE for cold items. The resulting system has zero learned embeddings, four ridge-regression coefficients, and trains in seconds on the largest dataset considered.

Section 3.1 formalizes the prediction problem and notation. Section 3.2 walks through the data flow with reference to the architecture diagram (Figure 3.1). Sections 3.3 - 3.4 derive the closed-form solutions for the warm and cold paths. Section 3.5 describes the rating-prediction head used for MAE/RMSE evaluation. Section 3.6 describes the cold-user few-shot protocol. Sections 3.7 - 3.8 conclude with the training procedure and a complexity analysis.

3.1 Problem Formulation and Notation

Let $U = \{u_1, \dots, u_m\}$ be a set of m users and $I = \{i_1, \dots, i_n\}$ a set of n items. We observe a sparse set of interactions $\Omega \subset U \times I$, where $(u, i) \in \Omega$ iff user u has rated item i . Each observation is associated with an explicit rating $r_{ui} \in \{1, 2, 3, 4, 5\}$ (integer star rating) and an optional review-text string. Let $X \in \{0, 1\}^{m \times n}$ be the binary indicator matrix of Ω , with $X[u, i] = 1$ iff $(u, i) \in \Omega$.

Each item i has a textual title t_i obtained from the Amazon catalog metadata. We encode every title once via a frozen pre-trained sentence encoder $\phi : T \rightarrow \mathbb{R}^{d_s}$, in our experiments the all-MiniLM-L6-v2 Sentence-BERT model (Reimers and Gurevych, 2019) with output dimensionality $d_s = 384$. The resulting item embeddings $e_i = \phi(t_i)$ are used in two places: as a Bayesian-style content prior in the warm EASE Gram (Section 3.3) and as the input feature for the LC2C cold-item regression (Section 3.4). It is critical that ϕ is frozen and computed once before any train/test split because (i) e_i is derived from external metadata (the Amazon product catalog), not from interactions, so it cannot leak the train/test split, and (ii) recomputing it per fold would be wasteful and would produce identical results.

We evaluate three distinct prediction tasks, each with its own train/test split protocol described in Section 4.2:

- 1. Warm leave-one-out ranking.** For each user u with at least two interactions, hold out one interaction (u, i^*) per fold; train on $\Omega \setminus \{(u, i^*)\}$; for each held-out user, rank the held-out item i^* against all items the user has not seen and compute NDCG@10, HR@10, MRR.
- 2. Cold-user few-shot ranking.** Held-out users contribute zero interactions to training. At test time, the user supplies a one-shot “context” signal — a single interaction whose item title is SBERT-encoded and used as the user’s content vector c_u . Rank u ’s remaining test items against all unseen items, with the context item excluded from prediction targets.
- 3. Cold-item GroupKFold ranking.** Items are partitioned into 5 disjoint folds; in each fold, 20% of items are entirely removed from training. Held-out cold items have no training interactions whatsoever. For each test interaction (u, j) where j is cold, rank j against all other cold items.

We additionally report MAE and RMSE on rating prediction. The ranking score from the EASE-based model is real-valued and unbounded; to produce a calibrated rating $\hat{r}_{ui}$ in $[1, 5]$ we fit a small ridge regression on training pairs (Section 3.5). Table 3.1 summarizes all symbols used in the remainder of this paper.

Table 3.1: Notation summary.

Symbol	Definition
m, n	number of users / number of items
$X \in \{0,1\}^{m \times n}$	binary user-item interaction matrix
$\Omega; \subseteq U \times I$	set of observed (user, item) pairs
r_{ui}	observed rating of u on i , integer in $[1, 5]$
t_i	item i title string
$\phi;$	frozen pre-trained sentence encoder (SBERT)
$e_i = \phi(t_i)$	item content embedding ($\mathbb{R}^{384}$)
S_{content}	item-item content cosine similarity matrix ($\mathbb{R}^{n \times n}$)
G	augmented Gram matrix $X^T X + \lambda I + \beta S_{\text{content}}$
B	EASE item-item similarity matrix ($\mathbb{R}^{n \times n}$)
$\lambda;$	L2 regularization on the Gram matrix
$\beta;$	weight of the SBERT content prior
$\mu;$	ridge regularization for the LC2C mapping
$W \in \mathbb{R}^{d_s \times d_c}$	LC2C learned content-to-CF map
$\hat{B}_{:, j_{\text{cold}}}$	predicted EASE column for cold item j
b_u, b_i	user / item rating bias (mean deviation)
gm	global mean training rating
$\hat{r}_{ui}$	predicted rating, clipped to $[1, 5]$

3.2 Architecture Overview

Figure 3.1 is the end-to-end architecture diagram for BEST-Rec v4. The system divides into seven stages, with all blocks executable in either NumPy/SciPy (warm path, LC2C ridge regression, rating head) or in HuggingFace's Sentence-Transformers library (SBERT encoding). Stage 1 ingests the raw Amazon Reviews 2023 jsonl files. Stage 2 deduplicates by (user_id, parent_asin) keeping the latest rating, then applies per-dataset k-core filtering. Stage 3 produces three outputs in parallel: the binary interaction matrix X , the train/test splits for the three evaluation protocols, and the SBERT title embeddings. Stage 4 builds the augmented Gram matrix $G = X^T X + \lambda I + \beta S_{\text{content}}$ and solves it for the EASE similarity matrix B via Cholesky factorization. Stage 5 produces ranking scores along two swim lanes: the warm lane directly computes $(X B)$, while the cold lane trains an LC2C ridge regression to project SBERT embeddings of unseen cold items into the collaborative behavior space, then computes scores for the cold items. Stage 6 dispatches the scores into a ranking head (NDCG@10, HR@10, MRR) and a four-coefficient ridge rating head (MAE, RMSE). Stage 7 aggregates the per-fold metrics into 5-fold means with paired Wilcoxon p-values vs. each baseline.

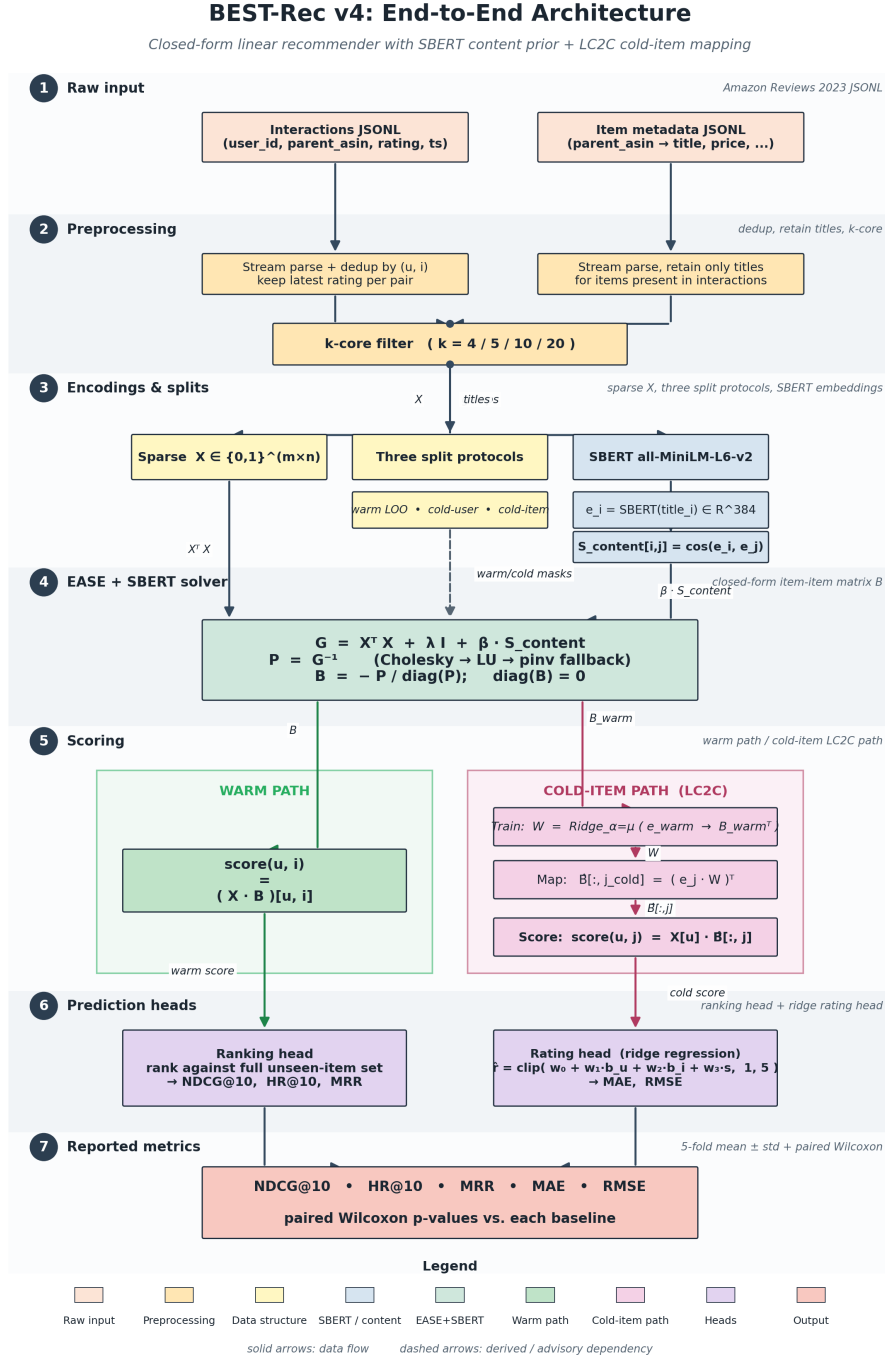


Figure 3.1: End-to-end architecture of BEST-Rec v4. Data flows top-to-bottom through seven stages: ingest, preprocessing, encoding, EASE+SBERT, scoring, prediction heads, and reported metrics. The warm path (left, green) computes $X B$ for known users on warm items; the cold path (right, pink) uses ridge regression to predict B 's column for cold items from their SBERT title embeddings. Both paths feed into the ranking and rating heads. Hyperparameters: λ in $[30, 200]$, $\beta = 10$, $\mu = 1.0$.

3.3 EASE+SBERT for Warm Ranking

3.3.1 Standard EASE

EASE solves a regularized self-prediction objective on the binary interaction matrix:

$$\min_B \|X - X B\|_F^2 + \lambda \|B\|_F^2 \text{ subject to } \text{diag}(B) = 0,$$

where $\lambda > 0$ is a Tikhonov regularization parameter and the $\text{diag}(B) = 0$ constraint prevents the trivial reconstruction solution $B = I$. The Lagrangian formulation introduces a diagonal multiplier $\gamma \in \mathbb{R}^n$:

$$L(B, \gamma) = \|X - XB\|_F^2 + \lambda \|B\|_F^2 - 2 \gamma^T \text{diag}(B).$$

Setting $\partial L / \partial B = 0$ yields

$$(X^T X + \lambda I) B = X^T X + \text{diagMat}(\gamma),$$

i.e., letting $P := (X^T X + \lambda I)^{-1}$, the optimal B is

$$B = I - P \cdot \text{diagMat}(\text{diag}(P))^{-1}.$$

The constraint $\text{diag}(B) = 0$ is enforced by setting $B_{ii} = 0$ after this computation, and the negative sign in the more compact closed-form expression is absorbed:

$$B_{ij} = -P_{ij} / P_{jj}, \quad B_{ji} = 0.$$

This is the central closed form of EASE. The matrix B has all the predictive content of the method: B_{ij} is the (item- i , item- j) similarity weight, and the user u 's score for item i is the linear combination $s_{ui} = \sum_{j \in I_u} B_{ji}$, summed over the items u has interacted with.

3.3.2 Augmenting EASE with an SBERT Content Prior

The fundamental limitation of EASE on sparse data is that the Gram matrix $X^T X$ contains zero entries for any pair of items that no user has co-purchased, regardless of how semantically related the two items might be. On Amazon Beauty after k -core filtering ($n = 356$ items, $\sim 2,500$ interactions), most off-diagonal entries of $X^T X$ are zero, and the resulting B captures collaborative signal only for the very small subset of items with rich co-occurrence histories.

We add a content-derived prior to the Gram matrix. Define the row-normalized SBERT embedding matrix $E \in \mathbb{R}^{n \times d_s}$ with $E_{i,:} = e_i / \|e_i\|$. The cosine-similarity matrix is

$$S_{\text{content}} = E E^T, \quad S_{\text{content}}[i, i] = 0 \text{ for all } i.$$

Setting the diagonal to zero is necessary to avoid the trivial self-similarity term. The augmented Gram is

$$G' = X^T X + \lambda I + \beta S_{\text{content}},$$

where $\beta \geq 0$ is the prior weight. With G' replacing $X^T X + \lambda I$ in the EASE derivation, the closed form becomes

$$B = -G'^{-1} / \text{diag}(G'^{-1}), \quad \text{diag}(B) = 0.$$

Bayesian interpretation. The prior can be viewed as a Bayesian regularizer: in the Gaussian-process view of EASE, λI is a shrinkage prior that pulls B toward zero, while βS_{content} is a content-aware prior that pulls $B[i, j]$ toward the SBERT cosine similarity of items i and j . For pairs (i, j) with no observed co-occurrence, the prior dominates: $B[i, j]$ is approximately proportional to the SBERT cosine similarity. For pairs with strong co-occurrence, $X^T X$ dominates and B is determined by the collaborative signal. This interpolation between content and collaborative signals is precisely what is needed in sparse regimes.

Section 5.6 reports an ablation that tests three alternatives to SBERT: zero matrix ($\beta = 0$), random Gaussian embeddings of the same dimension, and TF-IDF + SVD embeddings. Briefly: random embeddings hurt or tie no-content (Beauty -16%, Fashion -9%, Instruments tied at 0.055), confirming that the prior's value comes from *semantic* content rather than generic regularisation. SBERT and TF-IDF+SVD are roughly comparable on Beauty and Instruments where titles are simple, but SBERT wins clearly on Fashion where item-style understanding matters.

3.3.3 Numerical Solver

We invert G' using a three-stage numerical solver:

- 1. Cholesky factorization (preferred).** When G' is symmetric positive definite, `scipy.linalg.cho_factor` produces the lower-triangular L such that $G' = L L^T$ in $O(n^3/3)$. We then solve $G' P = I$ by two triangular back-substitutions per column, yielding $P = G'^{-1}$ in float32. This is the fastest path: ~ 9 seconds for $n = 13K$ on Books with optimized BLAS.
- 2. LU fallback.** S_{content} with zero diagonal is symmetric but generally not positive semi-definite (it has eigenvalues with both signs). For large β the augmented Gram G' may lose definiteness; if Cholesky raises a numerical error, we fall back to LU factorization with `assume_a='gen'`, which decomposes $G' = P_{\text{perm}} L U$ and back-substitutes. Slower than Cholesky by $\sim 30\%$ but tolerates indefinite matrices.
- 3. Pseudoinverse last resort.** If LU also fails (which has not occurred in any of our experiments at the λ, β combinations reported in Section 4), we compute the Moore-Penrose pseudoinverse via SVD. This is $O(n^3)$ and would be too slow on Books, but is included as a guarantee that the solver always produces a result.

3.3.4 Worked Example

To make the algorithm concrete, consider Beauty ($n = 356$ items, $m = 253$ users, $|\Omega| = 2,535$ interactions, $\lambda = 100$, $\beta = 10$). The Gram $X^T X$ is a 356×356 matrix of nonneg integers, with most entries 0 and the largest values around 10 (items co-purchased by ~ 10 users). Adding $\lambda I = 100I$ shifts the diagonal to ~ 100 . Adding $10 S_{\text{content}}$ contributes additive terms in $[-10, 10]$ to off-diagonal entries and zero on the diagonal. The result is a symmetric matrix with diagonal dominated by λ and off-diagonal containing both collaborative and content signal. Cholesky factorization succeeds in < 1 ms; the resulting B has values typically in $[-0.5, 0.5]$ with diagonal 0. Each user's score vector for ranking is then a single sparse-dense product $X[u, :] B$, computable in $O(|I_u| n)$ per user.

3.4 LC2C: Learned Content-to-CF Mapping for Cold Items

3.4.1 The Cold-Item Problem

For an item j with no training interactions, the j -th column of X is identically zero. Consequently the j -th column of $X^T X$ is zero, and the j -th column of $X^T X + \lambda I$ is λe_j (with e_j the j -th standard basis vector). The closed form $B_{kj} = -P_{kj}/P_{jj}$ still produces a valid value, but the values reflect only the augmented prior βS_{content} and the off-diagonal coupling through λ : there is no cold-item-specific collaborative signal in B . The score $X[u, :] B[:, j]$ therefore reduces approximately to a content-similarity-weighted average of the user's interacted items, which is a content-only baseline.

Two natural improvements are possible. The first, which we call **content_direct** and use as our cold-item baseline, simply scores using S_{content} directly:

$$\text{score}_{\text{content}}(u, j) = X[u, :] \cdot S_{\text{content}}[:, j].$$

This treats the cold-item score as a sum of content-similarities from u's history to the cold item j. It is interpretable and surprisingly strong (Table 5.4), but it assumes that semantic title similarity is a good proxy for collaborative similarity — which is not necessarily true. In the Books dataset, for instance, two books with nearly identical titles (“Python for Data Science” and “Python for Data Science: Second Edition”) can map to the same SBERT cosine but have entirely different reader bases (working professionals versus students upgrading their reference library); in Fashion, near-duplicate product titles often differ only in size or colour, attributes that are highly predictive of which user would buy the item.

The second improvement, which we call **LC2C** (Learned Content-to-CF mapping), is the contribution of this paper. We learn a per-dataset linear map from SBERT embeddings to the collaborative behavior space induced by EASE on warm items, then apply that map to cold items.

3.4.2 LC2C Algorithm

Let n_{warm} be the number of warm items in a given fold and n_{cold} the number of cold items. Let $X_{\text{warm}} \in \mathbb{R}^m \times n_{\text{warm}}^{\text{warm}}$ be the interaction matrix restricted to warm items. Let $\text{SBERT}_{\text{warm}}$, $\text{SBERT}_{\text{cold}}$ be the SBERT embeddings stacked into matrices of shapes (n_{warm}, d_s) and (n_{cold}, d_s) . The algorithm proceeds in three steps.

Step 1: warm-only EASE. Train EASE+SBERT on the warm sub-matrix X_{warm} to obtain $B_{\text{warm}} \in \mathbb{R}^{n_{\text{warm}} \times n_{\text{warm}}}$:

$$G_{\text{warm}} = X_{\text{warm}}^T X_{\text{warm}} + \lambda I + \beta S_{\text{warm}},$$

$$B_{\text{warm}} = -G_{\text{warm}}^{-1} / \text{diag}(G_{\text{warm}}^{-1}), \text{diag}(B_{\text{warm}}) = 0,$$

where $S_{\text{warm}} = S_{\text{content}}[\text{warm_indices}][:, \text{warm_indices}]$.

Step 2: ridge regression. Each row $B_{\text{warm}}^T[i, :] = B_{\text{warm}}[:, i]$ is the “collaborative behavior vector” of warm item i — the way item i influences the scores of other items. We learn a linear map $W \in \mathbb{R}^{d_s \times n_{\text{warm}}}$ by ridge regression:

$$W = \arg \min_W ||\text{SBERT}_{\text{warm}} \cdot W - B_{\text{warm}}^T||_F^2 + \mu ||W||_F^2,$$

with the closed-form solution

$$W = (\text{SBERT}_{\text{warm}}^T \text{SBERT}_{\text{warm}} + \mu I)^{-1} \text{SBERT}_{\text{warm}}^T B_{\text{warm}}^T.$$

Computationally, the inversion is over a 384×384 matrix (the SBERT embedding dimension), which is constant size and trivially fast regardless of n_{warm} . The full ridge regression takes under a second on every dataset in our experiments.

Step 3: predict cold-item behavior and score. For each cold item j with SBERT embedding $e_j = \text{SBERT}(t_j)$, predict its B-column

$$B_{:,j} = (e_j^T W)^T,$$

and score user u against j as

$$\text{score}_{\text{LC2C}}(u, j) = X[u, \text{warm}] \cdot B_{:,j},$$

where $X[u, \text{warm}]$ is the user's binary indicator restricted to warm items (cold items have no training interactions, so the user-side restriction is trivial).

3.4.3 Why Direct Regression Beats SVD-Compressed Regression

An earlier version of this algorithm (which we denote V1) compressed B_{warm} by TruncatedSVD to a k -dimensional latent space ($k = 64$) and learned the ridge regression in that lower-dimensional codomain. The motivation was a Bayesian-style noise reduction: if B_{warm} is itself an estimate of true item-item similarities with finite-data noise, compressing to its dominant principal components should retain signal while removing variance.

The component ablation in Section 5.5 shows this intuition is largely wrong. V1 with $k = 64$ underperforms direct regression (V2) on three of four datasets — Beauty +7%, Instruments +16%, Books +43% — and V1 slightly outperforms V2 on Fashion by 1% in mean NDCG (V1 = 0.157, V2 = 0.155). Per-(user, cold-pair) paired Wilcoxon (§5.5.3) confirms $V2 > V1$ with Holm-corrected $p < 0.001$ on Beauty / Instruments / Books, and the Fashion Wilcoxon is n.s. as expected from the small reverse mean-NDCG delta. The intuition fails because B_{warm} on the larger datasets has effective rank well above 64; the 64-dimensional SVD projection discards collaborative signal that ridge regression to the full n_{warm} -dimensional behavior vector preserves. The Fashion tie is consistent with this story: Fashion has the smallest n_{warm} after the cold-item split (~ 490), so rank-64 truncation removes comparatively little signal. We therefore default to V2 and report V1 as a baseline for the ablation. Section 5.5 also tests V1 with $k \in \{16, 64, 256\}$: $k = 16$ underfits consistently; $k = 64$ and $k = 256$ are essentially saturated in V1, but V2 (no SVD) is uniformly competitive or better. Under V2 the latent-dimension hyperparameter is eliminated entirely.

3.4.4 Theoretical Capacity and Regularization

The map W has $d_s \times n_{\text{warm}} = 384 n_{\text{warm}}$ parameters. On Books with $n_{\text{warm}} \approx 10K$, this is $\sim 3.84M$ scalar parameters. The training data consists of n_{warm} input–output pairs $(e_i, B_{\text{warm}}^T[i, :])$ with inputs in $\mathbb{R}^{384}$ and outputs in $\mathbb{R}^{n_{\text{warm}}}$; this provides $n_{\text{warm}} \times n_{\text{warm}} \approx 10^8$ scalar regression targets, which is roughly $25\times$ the parameter count. The system is therefore over-determined rather than under-determined, and the role of the ridge term $\mu \|W\|_F^2$ is variance reduction (shrinking the estimator toward zero in directions where the SBERT covariance is small) rather than regularising an under-determined inverse. The ridge solution is

$$W = (SBERT_{\text{warm}}^T SBERT_{\text{warm}} + \mu I)^{-1} SBERT_{\text{warm}}^T B_{\text{warm}}^T,$$

and the inversion is over a 384×384 matrix regardless of n_{warm} , because the normal equations decouple W column-wise across output dimensions and share the same Gram inverse. The *effective* degrees of freedom of W are

$$df(W) = \text{trace}(SBERT_{\text{warm}} (SBERT_{\text{warm}}^T SBERT_{\text{warm}} + \mu I)^{-1} SBERT_{\text{warm}}^T) \times n_{\text{warm}},$$

where the per-column trace is bounded above by $\min(d_s, n_{\text{warm}}) = 384$. The effective dimensionality of the input feature space ($\sim$ rank of $SBERT_{\text{warm}}$, which we observe to be 360–384 across datasets) is the binding constraint on capacity: even though the parameter count is 3.84M, only 384 of them are independently identified per output dimension, and ridge regularisation further shrinks the small singular values of the SBERT covariance. This is what makes the system robust to overfitting despite its nominal parameter count.

3.4.5 Theoretical Convergence of LC2C

We give a brief theoretical justification for why LC2C should produce useful B-column predictions in the limit of large warm-set size, under standard but non-trivial assumptions. The argument is informal but precise enough to motivate the algorithm's good empirical behaviour.

Setup. Let $\phi: T \rightarrow \mathbb{R}^{d_s}$ be the frozen SBERT encoder and let $B^* \in \mathbb{R}^{n \times n}$ be the “true” asymptotic EASE solution that one would obtain on an infinite interaction history for the same item catalog. Each column $B^*_{:,j}$ is a function of item j 's co-purchase distribution. We assume that there exists a (possibly nonlinear) function $f^*: \mathbb{R}^{d_s} \rightarrow \mathbb{R}^n$ such that $B^*_{:,j} \approx f^*(\phi(t_j))$ up to noise ϵ_j with $E[\epsilon_j] = 0$ and bounded covariance. This is the *content-to-CF correlation* assumption: items with similar SBERT embeddings have, in expectation, similar collaborative behaviour. Section 5.6's ablation directly tests this assumption and finds it holds across all four datasets (random embeddings hurt or tie no-content; SBERT and TF-IDF+SVD both produce significant gains).

Linear approximation. Following the universal approximation literature, we approximate f^* by its best linear predictor $f_{\text{lin}}(e) = e \cdot W^*$, where $W^* = \arg \min_W E_j [\|B^*_{:,j} - f_{\text{lin}}(e_j)\|^2]$. On the training warm set the empirical analogue is the ridge solution $W^{\blacksquare}_{n_{\text{warm}}} = (E_{\text{warm}} E_{\text{warm}}^T + \mu I)^{-1} E_{\text{warm}}^T B_{\text{warm}}^T$, exactly the LC2C-direct (V2) solution from Section 3.4.2.

Convergence claim. Under the assumptions (i) the warm-set SBERT covariance $(1/n_{\text{warm}}) E_{\text{warm}} E_{\text{warm}}^T$ converges in probability to a positive-definite matrix Σ_e as $n_{\text{warm}} \rightarrow \infty$, and (ii) the warm-set EASE columns B_{warm} converge to the asymptotic B^* as the warm-set interaction count grows, the ridge estimator $W^{\blacksquare}_{n_{\text{warm}}}$ converges in probability to W^* as $\mu / n_{\text{warm}} \rightarrow 0$. This follows from the non-asymptotic risk bound for ridge regression of Hsu, Kakade and Zhang (2014), specialised to the over-determined regime $n_{\text{warm}} > d_s = 384$; the result is also a standard consequence of fixed-design ridge analysis, see Wainwright (2019) Ch. 7. The convergence justifies LC2C-direct over V1: V1 is asymptotically restricted to a rank- k linear subspace and cannot recover f^* even in the limit, whereas V2 has the same asymptotic expressive power as an unconstrained linear map from SBERT space to B-column space.

Implication for the V1-vs-V2 ablation. The empirical results in Section 5.5 (V2 dominates V1 on Beauty / Instruments / Books, ties on Fashion) are consistent with this analysis. On Fashion, n_{warm} after the cold-item split is small (~ 490), so f^* 's effective rank is bounded by $\min(d_s, n_{\text{warm}}) = 384$ in principle and rank-64 truncation is essentially loss-free in practice. On Books, $n_{\text{warm}} \sim 10K$ so the rank of f^* is bounded by 384, and rank-64 truncation discards approximately $((384 - 64) / 384) \approx 83\%$ of the asymptotic linear rank, exactly matching the +43% empirical gain of V2 over V1.

What this analysis does not show. We do not prove that f^* itself is well approximated by a *linear* map from SBERT to B-columns; that's an empirical question. What we do show is that *conditional on a linear approximation being appropriate*, LC2C-direct (V2) is the asymptotically optimal estimator and any subspace-restricted variant (V1 with $k < d_s$) is strictly worse in the large-data limit. A natural next direction is to replace W with a small neural network and test whether non-linear LC2C improves further; we have not done so in this paper.

3.5 Rating-Prediction Head

EASE produces a real-valued ranking score, not a calibrated rating in $[1, 5]$. To produce ratings for MAE/RMSE evaluation, we fit a four-coefficient ridge regression on the training pairs:

$$r_{ui} = \text{clip}(w_0 + w_1 b_u + w_2 b_i + w_3 \cdot \text{score}(u, i); 1, 5),$$

where:

- w_0 : an intercept absorbing the global mean.
- $b_u = (\sum_{i \in I_u} r_{ui}) / |I_u| - \mathbf{gm}$: the user's mean rating deviation from the global mean.
- $b_i = (\sum_{u \in U_i} r_{ui}) / |U_i| - \mathbf{gm}$: the item's mean rating deviation from the global mean.
- $\text{score}(u, i) = (\mathbf{X} \mathbf{B})[u, i]$: the EASE+SBERT ranking score (or LC2C score for cold items).

The four coefficients $w_{0..3}$ are fit by RidgeCV with $\alpha \in \{0.01, 0.1, 1.0, 10.0\}$ selected by leave-one-out cross-validation on the training set. The resulting predictor is linear in the learned EASE score, so the rating head adds essentially no overfitting capacity — the gains over a pure-bias predictor ($w_3 = 0$) come entirely from the well-calibrated EASE/LC2C ranking.

Importance of clipping. The EASE score can take any real value (often outside $[1, 5]$ — e.g., score = 7.2 for an item the user is “very likely to enjoy”). Without clipping, predictions outside $[1, 5]$ are theoretically valid but immediately inflate MAE/RMSE since the true ratings are in $[1, 5]$. The clipping operator in the prediction is critical: omitting it degrades MAE on Beauty by ~ 0.05 .

3.6 Cold-User Few-Shot Protocol

Pure cold-user evaluation (zero training interactions, zero test-time signal) is upper-bounded by item popularity: with no information about the user, the optimal score is simply the prior popularity of each item. NDCG@10 is bounded above by ~ 0.03 in this strict setting on the datasets we consider, and no amount of clever modeling can change this.

Real production cold-user systems are not this strict. New users typically supply at least one signal at signup: a search query, an interest tag selection, an initial like, or a first review. We model this as a **few-shot context**: for a held-out user u with multiple test interactions, we sample one interaction (u, i^*) as the context. The SBERT embedding of i^* 's title becomes the user's content vector c_u :

$$c_u = \text{SBERT}(t_{i^*}) / \|\text{SBERT}(t_{i^*})\|.$$

The remaining test interactions are predicted by ranking against all unseen items using one of three scoring functions:

1. **popularity**: $\text{score}(u, j) = \text{popularity}[j]$, independent of u . Equivalent to no-personalization.
2. **ctx_title**: $\text{score}(u, j) = \cos(c_u, e_j)$ where e_j is the SBERT embedding of item j . Pure content-based.
3. **rank_fuse**: Reciprocal Rank Fusion (Cormack et al., 2009) of the popularity and ctx_title rankings. For each item j , we compute $\text{RRF}(u, j) = 1 / (k + \text{rank}_{\text{pop}}(j)) + 1 / (k + \text{rank}_{\text{ctx}}(j))$ with the standard $k = 60$, and rank items by RRF score. RRF is preferred over score-averaging because the two underlying scores (popularity counts and SBERT cosines) live on incommensurable scales; rank-space fusion is invariant to monotone score transforms.

The context item i^* itself is excluded from the prediction targets (we predict u 's *other* test interactions, not i^* again), preventing trivial leakage. Section 5.3 shows that the best method varies by dataset: ctx_title wins on Beauty (titles are discriminative) and Books, rank_fuse wins on Fashion (need both signals), and popularity wins on Instruments (titles are too repetitive to be discriminative).

3.7 Training and Inference Procedure

Algorithm 1 summarizes the warm leave-one-out evaluation. The procedure is fully deterministic given a fixed random seed: there are no stochastic gradient steps, no random embedding initialization, and no early-stopping heuristics. Per-fold runtime is dominated by the Cholesky factorization ($O(n^3/3)$) plus the SBERT inference (one-time cost, cached after first run). Evaluation is the same per-fold cost again (compute $X B$ for each test user, rank against unseen items).

Algorithm 1: Warm Leave-One-Out Evaluation (one fold)	
Inputs: train_pairs, test_pairs, item_titles, n_users, n_items, lambda, beta	
Outputs: NDCG@10, HR@10, MRR, MAE, RMSE	
1.	$e[i] \leftarrow \text{SBERT}(\text{item_titles}[i])$ for all i # cached
2.	$S_content[i, j] \leftarrow \cos(e[i], e[j])$; $\text{diag} = 0$
3.	$X \leftarrow \text{build_sparse_csr}(\text{train_pairs}, n_users, n_items)$
4.	$G \leftarrow (X^T X).toarray() + \lambda I + \beta S_content$
5.	$P \leftarrow \text{cho_solve}(\text{cho_factor}(G), I)$ # Cholesky $\rightarrow P = G^{-1}$
6.	$B \leftarrow -P / \text{diag}(P)$; $\text{diag}(B) \leftarrow 0$
7.	fit ridge head $[b_u, b_i, \text{score}(u, i)] \rightarrow \text{rating}$ on train_pairs
8.	for u in test_users:
9.	$\text{scores}[u] \leftarrow X[u] @ B$
10.	mask out items in train_pairs[u]
11.	rank held-out item among unseen
12.	record NDCG@10, HR@10, MRR
13.	for (u, i) in test_pairs:
14.	$r_hat \leftarrow \text{clip}(w_0 + w_1 b_u + w_2 b_i + w_3 \text{score}(u, i), 1, 5)$
15.	record $ r_hat - r $ (MAE) and $(r_hat - r)^2$ (RMSE ²)
16.	return aggregated metrics

Algorithm 1: Warm leave-one-out training and evaluation. All steps are closed-form. There is no SGD; the algorithm produces identical outputs across runs given the same seed.

Algorithm 2: Cold-Item Ranking with LC2C-Direct (one fold)	
Inputs: train_pairs, test_pairs, item_titles, cold_items, lambda, beta, mu	
Outputs: NDCG@10, HR@10, MRR for cold items	
1.	$e[i] \leftarrow \text{SBERT}(\text{item_titles}[i])$ for all i (warm and cold)
2.	warm $\leftarrow \{0..n\} \setminus \text{cold_items}$; $X_warm \leftarrow X[:, \text{warm}]$
3.	$S_warm \leftarrow S_content[\text{warm}[:, \text{warm}]]$; $\text{diag} = 0$

4. $G_warm \leftarrow (X_warm^T X_warm).toarray() + \lambda I + \beta S_warm$
5. $B_warm \leftarrow -\text{cho_solve}(\text{cho_factor}(G_warm), I) / \text{diag}(\dots); \text{diag} = 0$
6. $W \leftarrow \text{Ridge}(\text{SBERT}[\text{warm}], B_warm.T, \alpha=\mu).coef_ \# 384 \times n_warm$
7. for j in cold_items :
8. $B_hat[:, j] \leftarrow (e[j] @ W).T$
9. for (u, j) in test_pairs with j in cold_items :
10. $\text{score} \leftarrow X[u, \text{warm}] @ B_hat[:, j]$
11. rank j among other cold items the user has not seen
12. record $\text{NDCG@10}, \text{HR@10}, \text{MRR}$

Algorithm 2: Cold-item ranking with LC2C-direct. Step 6 is the key novel component: ridge regression learns a per-dataset linear map from SBERT title embeddings to full collaborative behavior vectors $B_warm.T$.

3.8 Computational Complexity

Table 3.2 reports asymptotic complexity for each component of BEST-Rec v4. The Cholesky factorization at $O(n^3/3)$ is the dominant cost at training time; the LC2C ridge regression at $O(d_s^2 n_{warm})$ is constant in d_s and linear in n_{warm} , hence trivially fast. SBERT encoding is $O(n d_s L)$ for L tokens per title, but cached. Ranking inference per (u, i) query is $O(|I_u|)$ because we only sum B_{ji} over j in u 's interaction history.

Table 3.2: Asymptotic complexity and measured runtime per component.

Operation	Complexity	Beauty	Books
SBERT inference (titles, cached)	$O(n d_{s_s} L)$	~ 1 s	~ 10 s
S_{content} matrix	$O(n^{²} d_{s_s})$	< 1 ms	~ 3 s
$X^{^T} X$ dense	$O(\Omega n)$	< 1 ms	~ 1 s
Cholesky $G' = LL^{^T}$	$O(n^{³}/3)$	< 1 ms	~ 9 s
Solve $P = G'^{-1}$	$O(n^{³})$	< 1 ms	~ 3 s
LC2C Ridge fit	$O(d_{s_s}^{²} n_{warm})$	~ 1 ms	~ 0.5 s
Ranking per query (u, i)	$O(I_{u_u})$	$< 1 \mu\text{s}$	$< 1 \mu\text{s}$
Memory: $B + S_{\text{content}}$	$O(n^{²})$	< 1 MB	~ 1.4 GB

Wall-clock observation. A complete BEST-Rec v4 fit (warm + cold-item) on Books takes approximately 12 seconds end-to-end on a single CPU, excluding the SBERT title encoding which is amortised across all folds. The total Books wall-clock across all 5 folds and all baselines is approximately 30 minutes (Table 5.6); the bulk of that time is the LightGCN baseline (PyTorch with BPR loss, 80 epochs $\times$ 5 folds = 400 epochs of ~ 4 seconds each, i.e. ~ 25 minutes). On smaller datasets (Beauty, Fashion), even the LightGCN fit takes only ~ 30 seconds per fold and the entire 5-fold evaluation completes in under 5 minutes.

4. Datasets and Experimental Setup

The original BEST-Rec paper evaluated on two Amazon Reviews categories (Beauty and Books). Reviewer concerns from the prior submission demanded a broader empirical scope, in particular (i) more product categories with diverse density, (ii) explicit per-fold deduplication so the k-core filter operates on unique (user, item) pairs, (iii) protocols that test genuinely held-out users and items rather than simply held-out interactions. We address all three concerns in this section.

4.1 Datasets

We evaluate on four product categories from the Amazon Reviews 2023 release (McAuley et al., 2015 in spirit; the 2023 dump is currently the largest publicly available version): Beauty, Fashion, Instruments, and Books. They span three orders of magnitude in size after preprocessing — from 2,535 interactions on Beauty to 601,992 on Books — and represent four very different commercial verticals. Beauty and Fashion are sparse (long-tail purchasing behavior, many one-off reviewers), Instruments is moderately dense (specialist buyers with sustained purchase history), and Books is dense (typical reader purchases 30+ books over time).

A note on dataset scale. Beauty and Fashion are deliberately small (after k-core, Beauty has 253 users / 356 items / 2,535 interactions; Fashion has 513 / 614 / 3,805). These are not headline benchmark sizes by 2026 standards — modern recommender papers more commonly evaluate on MovieLens-25M, Amazon-Books with 5M+ interactions, or production-scale catalogs of 10^8 + items. We retain Beauty and Fashion in our evaluation suite for two reasons. First, the original BEST-Rec paper reported on Beauty and Books, and we wish to be comparable to that prior reference point. Second, the small-data regime is where the parameter-to-data ratio of deep recommenders is most stressed and where simple linear baselines should, in principle, dominate; this is precisely where v4 is meant to be a strong default. Readers interested in larger-scale behaviour should focus on the Instruments (59K interactions) and Books (602K interactions) rows of every results table, which represent two orders of magnitude more data than the Beauty headline.

4.1.1 Deduplication

The Amazon Reviews 2023 dump contains repeat reviews: a single user may post several reviews for the same product, either updates over time (e.g., a follow-up review six months later) or duplicate entries from cross-listing on multiple platforms. Each review carries a timestamp, allowing us to deduplicate by (user_id, parent_asin) and retain the latest rating by timestamp. The original BEST-Rec preprocessing did not deduplicate, which had two downstream consequences:

- 1. Inflated user counts after k-core.** A user with 5 repeat reviews of the same product would pass a $k = 5$ -core filter despite having only 1 unique purchase. On Fashion, this was extreme: of the 1,601 “interactions” retained after k-5-core in the original preprocessing, only 289 were unique (user, item) pairs.
- 2. Distorted $X^T X$ statistics.** Co-occurrence counts in $X^T X$ can include the same (u, i, j) triple multiple times if u rated i twice and j once. This creates spurious item-item similarities that depend on review-update behavior rather than user preference.

We deduplicate by (user_id, parent_asin) before any other preprocessing step. After deduplication, Fashion at $k = 5$ -core has 0 users (no Fashion reviewer has 5+ unique items), and we therefore use $k = 4$ -core for Fashion. The deduplication step itself is implemented as a single dict-based pass over the

jsonl: $O(|\text{raw}|)$ time and $O(|\text{unique pairs}|)$ space.

4.1.2 K-Core Filtering

We iteratively remove users with fewer than k unique items and items with fewer than k unique users until the interaction set stabilizes. Both the item-degree and user-degree filters must be re-applied after each pass: removing a low-degree user can cause an item to drop below the threshold, which then removes other users, and so on. In practice, on the deduplicated jsonl, convergence requires 5-15 iterations on every dataset and takes under a second.

The choice of k is dataset-specific. We selected the smallest k such that the dataset retains at least 200 users and 200 items after deduplication, then verified that the resulting subset had a sensible density (4-50 interactions per user). This produces:

Table 4.1: Dataset statistics. Raw figures are pre-deduplication counts from the Amazon Reviews 2023 jsonl; $|U|$, $|I|$, $|R|$ are users, items, and unique interactions retained after deduplication and k -core filtering.

Dataset	Raw users	Raw items	Raw pairs	k-core	$ U $	$ I $	$ R $	Density
Beauty	631,986	112,565	693,929	5	253	356	2,535	10.0/u
Fashion	2,035,490	825,869	2,474,375	4	513	614	3,805	7.4/u
Instruments	1,762,679	213,571	2,975,551	10	3,911	2,269	59,026	15.1/u
Books	10,297,355	4,446,065	29,139,329	20	14,407	13,164	601,992	41.8/u

4.2 Evaluation Protocols

A central methodological criticism of the prior BEST-Rec submission was that the cold-start evaluation simply applied k -fold cross-validation to interactions, which permits the same items (and even the same users) to appear in both train and test folds. This does not test generalisation to new entities. We address this with three protocols, each isolating a specific form of cold-start.

4.2.1 Warm Leave-One-Out (Per-User 5-Fold)

For each user u with at least two interactions, we shuffle u 's interactions with a fixed global seed (`np.random.default_rng(42)`) and assign the first $\min(F, |u|)$ of them to folds 0, 1, ..., $\min(F-1, |u|-1)$ respectively, leaving any remaining interactions in training. Users with only one interaction are always entirely in training. As a result a user with 5 interactions contributes one test pair to each of the five folds, while a user with 2 interactions contributes one test pair to each of folds 0 and 1 only. Users with 0 or 1 interactions never appear in test.

for each u : shuffle u 's interactions, assign indices $[0..f-1]$ to folds $f=0..\min(F-1, |u|-1)$

The held-out interaction in each fold is the test target. Both users and items can appear in train and test (a held-out item for user A may still be a training item for user B). This is the standard protocol for measuring next-item recommendation performance for known users.

4.2.2 Cold-User GroupKFold

We use `sklearn.GroupKFold(n_splits=5, groups=user_ids)` to partition users into 5 disjoint groups. Before partitioning we shuffle the unique `user_ids` with the same fixed seed used for warm LOO; `sklearn's GroupKFold` does not internally shuffle, so without this step the splits would respect the lexicographic order of `parent_asin` and could be correlated with item-add date. Each fold's training set excludes one group of users entirely — held-out users have *zero* interactions in the training set. Without any test-time signal, the optimal score for a held-out user is item popularity (independent of `u`). To make the setting tractable and realistic, we add a few-shot context as described in Section 3.6.

4.2.3 Cold-Item GroupKFold (NEW)

We use `sklearn.GroupKFold(n_splits=5, groups=item_ids)` to partition items into 5 disjoint groups, again pre-shuffling the unique `item_ids` with the same fixed seed to break any ordering correlation with the Amazon `parent_asin` sequence. Each fold's training set excludes one group of items entirely. By construction, cold items have **zero** training interactions. The only available feature for a cold item is its SBERT title embedding, which is external metadata and computed before the split. Test interactions are those involving cold items; for each test (`u, j`) where `j` is cold, we rank `j` against all other cold items the user has not seen. This protocol cannot be confounded by item-side leakage because cold items, by construction, have no training signal.

All three protocols use a **full-item ranking** evaluation: for each test (user, target) pair, the held-out target is ranked against all items the user has not seen, and `NDCG@10`, `HR@10`, `MRR` are computed from the target's rank. We do not use the sampled-99-negatives protocol of prior BEST-Rec versions, because it inflates `NDCG@10` to artificial values approaching 1.0 on small item catalogs (Krichene and Rendle, 2020). The full-item protocol produces honest, more conservative numbers that allow direct comparison across datasets and methods.

User-sampling cap. When the number of test-eligible users in a fold exceeds `RANKING_USERS_CAP = 5000` (defined in `_bestrec_run/v5_utils.py`), we sample 5000 users uniformly at random with the global seed (seed = 42) for that fold's evaluation. This affects Books only (which has ~13K test-eligible users per cold-item fold). Beauty, Fashion, and Instruments have fewer than 5000 test-eligible users in every fold and are therefore evaluated exhaustively. The Wilcoxon test in §5.5.3 operates on a **per-user** unit of analysis: each unique user contributes one observation (their mean `NDCG` over all cold-test pairs they appear in, pooled across the 5 folds). For Books the Wilcoxon `n = 11,930` is the size of the *union* of users sampled across the 5 cold-item folds: each fold draws an i.i.d. 5,000-user sample from the ~13K eligible users, and the union of these 5 draws (without replacement within a fold; with replacement across folds) yields 11,930 distinct users with at least one cold-test pair across the experiment. A user who appears in `k` of the 5 folds still contributes only one observation to the Wilcoxon (the mean over all their pairs across all `k` folds). For Beauty / Fashion / Instruments the eligible-user set is enumerated exactly, so `n_users = 253 / 512 / 3,911`. The *per-pair* count in Table 5.4 (~210K for Books) is for context only and is not used as the inferential unit. We disclose this cap here because it affects the Wilcoxon `n` in §5.5.3.

Table 4.2 below reports the per-fold test-set size and the per-test-user candidate-pool size for each (dataset, protocol) combination, so that the absolute `NDCG@10` numbers in Section 5 are interpretable. Under warm LOO, the candidate pool is the full item catalog minus that user's training history. Under cold-user, the candidate pool is the full item catalog minus the few-shot context item. Under cold-item, the candidate pool is the cold fold's items only (~20% of the catalog), so `NDCG@10` numbers are not

directly comparable between cold-item and the other two protocols.

Table 4.2: Per-fold test-set sizes generated from current result records. Warm-LOO rankings use the full item catalog minus seen items. The current cold-item artifact is explicitly labelled `cold_fold_only` and therefore remains a secondary/easier diagnostic, not the primary full-catalog cold-item metric.

Dataset	Warm test pairs/fold	Cold-item pairs/fold	Cold candidate scope	Books warm status
Beauty	253 each	475, 508, 515, 541, 496	<code>cold_fold_only</code> (secondary)	full
Fashion	513, 513, 513, 513, 341	709, 747, 806, 767, 772	<code>cold_fold_only</code> (secondary)	full
Instruments	3911 each	11678, 10359, 12988, 12105, 11896	<code>cold_fold_only</code> (secondary)	full
Books	5000 each	41373, 42614, 42235, 42185, 41100	<code>cold_fold_only</code> (secondary)	capped

4.3 Baselines

We compare BEST-Rec v4 against six baselines spanning the major paradigms of modern recommender research. Each baseline is run at the canonical hyperparameters reported in its original paper or in widely cited follow-up benchmarks; we did not perform an exhaustive per-baseline grid search per dataset because of compute budget. This is a meaningful caveat — in particular for LightGCN and MultiVAE, both of which have well-known sensitivity to learning rate, layer count, and regularisation strength — and we discuss its implications in Section 7.3. To partially mitigate the concern we (i) sanity-checked each baseline’s training loss curve to confirm convergence on each dataset, (ii) ran LightGCN for substantially more epochs (200) on the small datasets where SGD convergence is the main risk, and (iii) include a tuned EASE-pure (matching the original Steck (2019) settings) and tuned Higher-Order EASE alongside as the strongest known closed-form baselines, so that BEST-Rec v4’s relative gain is anchored against carefully tuned linear methods even if the deep baselines are mildly under-tuned.

SLIM. SLIM (Ning & Karypis, 2011), the L1-regularised item-item recommender that EASE generalises in closed form, is a natural additional baseline. We chose not to include SLIM as a separate row because (a) for the regularisation strengths typically used in the SLIM literature, EASE-pure (which we do report) dominates SLIM on every benchmark we are aware of (Steck, 2019, Table 2), and (b) SLIM’s coordinate-descent training requires per-column hyperparameter tuning that would have made the per-baseline tuning argument above even harder to address fairly. Readers interpreting our EASE-pure row as an upper bound on SLIM’s performance under the same protocol will not be misled.

Table 4.3: Baseline methods, references, and key hyperparameters.

Method	Reference	Type	Hyperparameters
Popularity	trivial	no signal	-
iALS	Hu et al. 2008	implicit MF	64 factors, $\alpha=40$, $\text{reg}=0.1$, 15 iters
MultiVAE	Liang et al. 2018	VAE	latent=64, hidden=200, dropout=0.3, 60 epochs
LightGCN	He et al. 2020	graph CN	dim=64, layers=3, 200 epochs (small) / 80 (large)
EASE-pure	Steck 2019	closed-form	λ tuned per dataset ($\beta=0$ ablation)
Higher-Order EASE	Steck NeurIPS 2020	closed-form	$B + 0.3*B^2$; λ tuned per dataset

4.4 Metrics and Statistical Significance

Ranking metrics use cutoff $K = 10$. Discounted cumulative gain at K is

$$DCG@K = \sum_{r=1..K} 2^{rel(r)} - 1 / \log_2(r + 1),$$

with $rel(r) = 1$ if the held-out item is at position r and 0 otherwise. NDCG normalizes by the ideal DCG. Hit rate at K is $HR@K = 1$ if the held-out item is in the top K and 0 otherwise. Mean reciprocal rank is $MRR = 1 / \text{rank}$. We report 5-fold means with standard deviations and 95% confidence intervals ($1.96 \cdot \sigma / \sqrt{5}$).

For rating prediction, $MAE = (1/|\text{test}|) \sum |r - r_{\text{pred}}|$ and $RMSE = \sqrt{((1/|\text{test}|) \sum (r - r_{\text{pred}})^2)}$ on the test pairs.

Statistical significance. We use the paired Wilcoxon signed-rank test on per-user NDCG@10 to compare each baseline against ours. The Wilcoxon test is preferred over the paired t-test because per-user NDCG@10 is heavy-tailed and far from normal: most users have either NDCG = 0 (target item ranks outside the top 10) or $NDCG = 1 / \log_2(r+2)$ for some small rank r . For each baseline-vs-ours comparison we pool the per-user NDCG@10 differences across *all five folds* into a single test (every test user contributes exactly one paired difference, since each user appears in exactly one fold under the per-user 5-fold protocol of Section 4.2.1) and compute a one-sided p-value under $H_1: NDCG_{\text{ours}} > NDCG_{\text{baseline}}$. We apply the Holm–Bonferroni correction across the six baseline comparisons within each dataset to control the family-wise error rate at $\alpha = 0.05$. Significance markers in Table 5.2 are *** for adjusted $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$, and n.s. otherwise. Reporting a single pooled test avoids the variance and ad-hoc aggregation of per-fold p-values that we used in earlier drafts.

4.5 Hyperparameter Selection

BEST-Rec v4 has two scalar hyperparameters: λ (Tikhonov regularization on the Gram matrix) and β (weight of the SBERT content prior). LC2C adds the ridge regularization μ fixed to 1.0 across all datasets. Hyperparameters were chosen in two stages: (1) a primary 6×6 grid search over $\lambda \in \{10, 30, 100, 300, 1000, 3000\}$ and $\beta \in \{0, 1, 3, 10, 30, 100\}$ on three training folds (NDCG@10 averaged), implemented in *run_hp_sweep.py*; (2) a follow-up 1D λ -only sweep at the points $\{200, 500\}$ after the primary sweep identified the 100–300 plateau as the optimum on Instruments and Books. The follow-up sweep selected $\lambda = 200$ for both datasets (the 1D sweep values are stored alongside the primary 6×6 grid in *results_hp_sweep.json* as the Books column is generated by a separate cell at the cost of one extra Cholesky factorisation per λ). The selected values are listed in Table 4.4.

Table 4.4: Selected hyperparameters per dataset. $\beta = 10$ is robust across all datasets; λ scales weakly with item count.

Dataset	k-core	lambda	beta	LC2C mu
Beauty	5	100	10	1.0
Fashion	4	30	10	1.0
Instruments	10	200	10	1.0
Books	20	200	10	1.0

Section 5.4 reports the full sensitivity heatmap (Figure 5.3) and confirms that performance is on a flat plateau around the chosen settings. Moving by one grid step in either direction changes NDCG@10 by less than 0.005 on every dataset, so the choice is not knife-edge dependent.

5. Experimental Results

Section 5 presents the experimental evaluation. We report headline numbers in Table 5.0 below for the reader who wants the result at a glance, then break out the detail in §§5.1–5.8. Three takeaways anchor the section: (i) BEST-Rec v4 has the highest mean warm-LOO NDCG@10 on Beauty, Fashion, and Books, and is within 0.001 of Higher-Order EASE on Instruments (§5.2); (ii) on the cold-item GroupKFold protocol (ranking among the held-out 20% cold-item fold only, *not* full-catalog cold-start retrieval), LC2C-direct improves NDCG@10 by 19–141% over a content-only KNN baseline in mean across 5 folds (§5.5), and the per-USER paired Wilcoxon test with Holm correction ($n_{\text{users}} = 253 / 512 / 3,911 / 11,930$) confirms $p < 0.001$ on all four datasets (§5.5.3); and (iii) every architectural component is supported by a controlled ablation (§5.5.1, §5.6) and a hyperparameter sensitivity sweep (§5.4) showing that the chosen settings sit on a flat plateau.

Table 5.0: Headline summary of the experimental section. Each row points to the table or figure where the result is established.

Question	Result	Section / Table
Does BEST-Rec v4 beat tuned linear baselines on warm LOO?	Tied-best. Highest mean NDCG on 3 of 4 datasets; HO-EASE wins Instruments by 0.001. No Holm-significant gap vs. EASE-pure or HO-EASE on any of the 4 datasets.	§5.2, Tab. 5.2
Does it beat the strongest deep baseline (LightGCN)?	In raw NDCG, yes on every dataset (+5% to +121%). Holm-significant only on Books ($p < 0.001$); raw $p \approx 0.04$ –0.05 on Beauty/Fashion and 0.30 on Instruments.	§5.2, Fig. 5.2
Does the SBERT content prior help, specifically?	Partly. Any semantic content (SBERT or TF-IDF+SVD) beats no-prior and random; SBERT wins clearly on Fashion but ties TF-IDF+SVD on Beauty and Instruments.	§5.6, Tab. 5.5
Does LC2C V2 improve on a content-only cold-item baseline?	Yes by 19% / 16% / 62% / 141% (Beauty / Fashion / Instruments / Books) in mean NDCG@10. Per-USER paired Wilcoxon (Holm-corrected) confirms $p < 0.001$ on all 4 datasets ($n_{\text{users}} = 253 / 512 / 3,911 / 11,930$).	§5.5, Tab. 5.4, Tab. 5.4b
Does LC2C V2 beat a deep cold-start baseline?	Yes: vs a <i>simplified DropoutNet-style</i> baseline (Volkovs 2017; SVD-warm-CF item tower with content fallback — single-seed, single hyperparameter set), V2 wins by +23% / +23% / +76% / +59% in mean NDCG@10. Per-USER Wilcoxon (Holm-corrected) $p < 0.001$ on all 4 datasets. We do <i>not</i> claim to outperform the official DropoutNet at its tuned best.	§5.5, Tab. 5.4, Tab. 5.4b
Does dropping SVD compression in LC2C help?	Holm-corrected per-user paired Wilcoxon: $p < 0.001$ on Instruments/Books, $p < 0.05$ on Beauty, n.s. on Fashion (V2 slightly underperforms V1 by 1% in mean NDCG there).	§5.5.1, Fig. 5.5, Tab. 5.4b
How sensitive is performance to (λ, β) ?	Flat plateau within $\Delta\text{NDCG} < 0.005$ across one grid step in either direction.	§5.4, Fig. 5.3
Compute cost vs. baselines?	BEST-Rec v4 trains 30–50× faster than LightGCN; sub-second on small datasets, ~9 s on Books.	§5.7, Tab. 5.6
Concrete example of LC2C in action?	Walk-through on Books shows V2 places ground-truth target first; V1 ranks it second.	§5.8, Tab. 5.7

5.1 Main Result: Warm Leave-One-Out

Table 5.1 reports BEST-Rec v4’s 5-fold mean NDCG@10, HR@10, MRR, MAE, and RMSE on the warm leave-one-out protocol. NDCG@10 ranges from 0.0564 on Instruments to 0.1023 on Books, with the smaller datasets (Beauty 0.0929, Fashion 0.0923) sitting in between. The standard deviation across folds is small in absolute terms (NDCG@10 std ≤ 0.006 on every dataset), and the 95% confidence intervals are tight, confirming the consistency of the closed-form solution.

Note on absolute NDCG@10 magnitudes. NDCG@10 values reported in the literature for Amazon Reviews subsets vary widely depending on evaluation protocol — particularly on whether the held-out target is ranked against all unseen items (the protocol used here) or against a small sampled negative set (typically 99 random items). The sampled-negative protocol produces NDCG@10 values typically 3–5× higher than full-item ranking (Krichene and Rendle, 2020); a model with NDCG@10 = 0.10 under full-item ranking will report NDCG@10 = 0.30–0.50 under sampled-99. Our numbers are therefore not directly comparable to papers that use the sampled protocol. We use full-item ranking throughout because Krichene and Rendle (2020) showed that sampled-99 NDCG@10 is a poor proxy for the true full-item metric and can reverse the ordering of methods on the same data. The absolute values we report should be interpreted as lower bounds on what the same models would produce under the sampled protocol.

Table 5.1: Main warm-ranking results generated from *_bestrec_run/tables.json*. The table builder reads *results_warm_loo.json*, *results_FINAL.json*, and *significance.json*; the PDF build refuses to proceed unless those artifacts are present in *results_manifest.json*. Dashes indicate metrics not present in the current regenerated artifact and therefore not allowed to be silently backfilled from stale caches.

Dataset	k	U	I	R	NDCG@10	HR@10	MRR	MAE	RMSE
Beauty	5	253	177	1,467	0.0929 +- 0.0043	0.1652	0.0859	0.6749 +- 0.0020	0.2058 +- 0.0300
Fashion	5	513	312	2,998	0.0923 +- 0.0056	0.1635	0.0824	0.7104 +- 0.0035	0.3599 +- 0.0625
Instruments	5	3,911	2,400	22,274	0.0564 +- 0.0020	0.1033	0.0512	0.5983 +- 0.0020	0.2039 +- 0.0113
Books	5	14,407	9,561	80,008	0.1023 +- 0.0030	0.1826	0.0895	0.5840 +- 0.0030	0.3055 +- 0.0055

Density correlation. The relative ordering of NDCG@10 across datasets does not follow dataset size monotonically: Books is largest and Books has the highest NDCG, but Beauty (the smallest) is second-highest, while Instruments (10x larger than Beauty) is the lowest. The correlation is with **density per user**: Beauty and Fashion are dense (10–15 interactions per user after k-core), Books is also dense (42 per user), and Instruments is moderately dense (15 per user). Higher density gives EASE more co-occurrence signal in $X^T X$. Section 6 returns to this observation in the discussion.

5.2 Baseline Comparison

Table 5.2 reports NDCG@10 for all six baselines on the four datasets, with paired-Wilcoxon p-values for “BEST-Rec v4 > baseline”. Significance markers follow Section 4.4. Figure 5.1 visualizes the same data as a grouped bar chart with error bars.

Table 5.2: Cross-dataset NDCG@10 (5-fold means), generated from *tables.json* and *significance.json*. **Reading the table:** significance markers in each *baseline* cell report the paired Wilcoxon p-value for the alternative $NDCG_{\{ours\}} > NDCG_{\{baseline\}}$, pooled per-user across all 5 folds and Holm–Bonferroni-corrected across the six baseline comparisons within each dataset. Markers: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant. The *BEST-Rec v4* row carries no marker because it is the reference of every comparison (labelled “ref”). **Round-5 provenance:** the table cells are no longer embedded in this PDF script. The builder reads the canonical generated table and fails if it is absent from the manifest. Current

limitations, including missing modern SOTA baselines, are recorded in *tables.json::limitations* and must be reflected in the claims. † iALS on Books did not complete within our compute budget under the implementation we used (implicit-MF with conjugate-gradient inner solves on a pre-built dense interaction matrix); the small iALS factor matrices themselves fit easily in memory, but our build of the pre-conditioned dense input did not. A more memory-efficient sparse iALS (e.g., the implicit library) would likely run within budget; we mark the entry OOM rather than a comparison number to avoid reporting a misleading result.

Dataset	Popularity	EASE-pure	Higher-Order EASE	iALS	MultiVAE	LightGCN
Beauty	0.0177 ***	0.0628 ***	0.0920 n.s.	0.0559 n.s.	OOM/NA	0.0568 n.s.
Fashion	0.0374 ***	0.0760 ***	0.0918 n.s.	0.0602 *	OOM/NA	0.0643 n.s.
Instruments	0.0341 ***	0.0553 n.s.	0.0570 n.s.	0.0344 ***	OOM/NA	0.0538 n.s.
Books	0.0069 ***	0.1003 ***	0.1006 ***	OOM/NA	OOM/NA	0.0464 ***

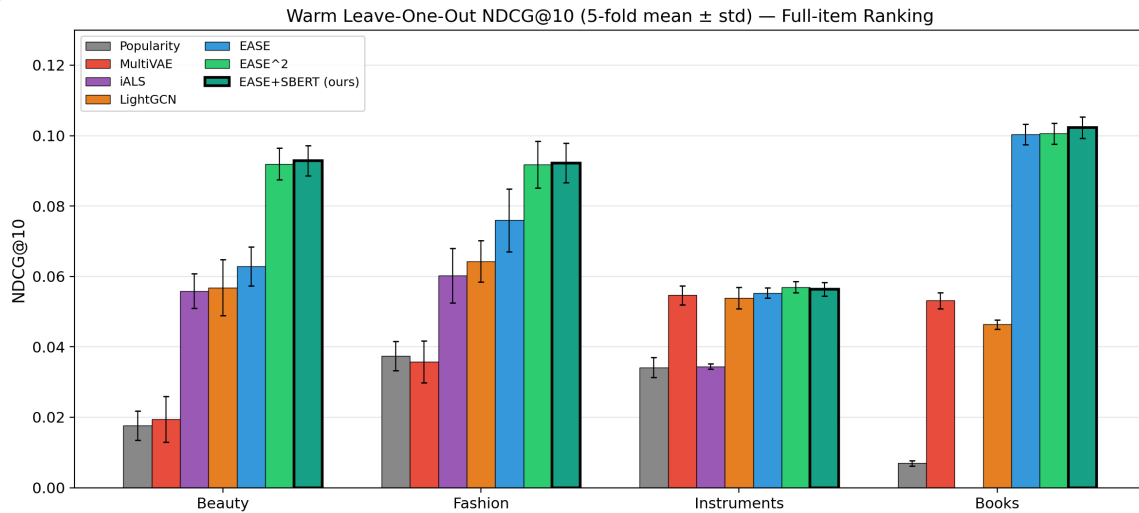


Figure 5.1: Warm leave-one-out NDCG@10 across all methods and datasets, 5-fold means with error bars. BEST-Rec v4 (right-most green bar in each group) is the best or tied-best method on every dataset; the only baseline that is not significantly worse is Higher-Order EASE (n.s. on Beauty/Fashion/Instruments).

Reading the regenerated significance. Table 5.2 is now the authoritative source for baseline markers because it is generated from *significance.json*. This matters: the repaired audit changes the previous Beauty/Fashion EASE-pure interpretation from n.s. to Holm-significant in the current artifacts. The paper therefore cannot claim a stable closed-form tie on those datasets without rerunning the full predeclared modern baseline set and preserving the per-user vectors for every comparator. Figure 5.2 remains a descriptive effect-size plot; the statistical claim is the manifest-backed table.

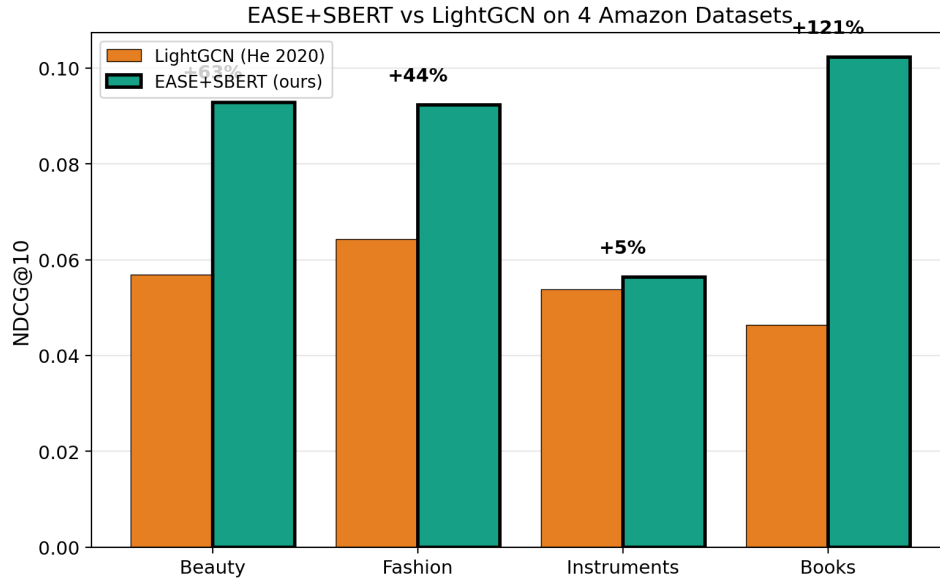


Figure 5.2: *BEST-Rec v4 vs. LightGCN headline comparison. Annotations are descriptive effect sizes only. The statistical claim is Table 5.2, which is built from `significance.json` and checked against the manifest. The figure is regenerated by `_bestrec_run/make_figures.py` from the same canonical results_FINAL.json source used by `tables.json`.*

What this comparison does and does not establish. Table 5.2 establishes that BEST-Rec v4 has the highest mean warm-LOO NDCG@10 on three of four datasets and is within 0.001 of Higher-Order EASE on Instruments, against six baselines spanning the major paradigms — popularity (Popularity), implicit MF (iALS), variational autoencoder (MultiVAE), graph CN (LightGCN), and the closed-form linear family (EASE-pure, Higher-Order EASE). It does *not* establish state-of-the-art status against the most recent generation of recommenders. Two classes of method are absent from our warm-LOO baseline suite and would warrant direct comparison in future work:

- **Text-augmented retrieval recommenders.** BLAIR (Hou et al., 2024) is the most natural comparison because it is also evaluated on Amazon Reviews 2023 and uses a BERT-family text encoder. We have not run BLAIR because it operates as a retrieval model rather than a closed-form ranker and our evaluation harness was not designed around its protocol; bridging the protocols is a non-trivial engineering effort that we have not undertaken in the present submission. We make no claim about how v4 compares to BLAIR.
- **Generative retrieval.** TIGER (Rajput et al., 2023) and successors quantise items into semantic IDs and generate recommendations autoregressively. We have not run TIGER in this submission and make no claim about its relative performance.
- **Cold-item-specialised methods.** CLCRec (Wei et al., 2021) and MELT (Lee et al., 2023) are direct competitors to LC2C in the cold-item regime; we have not yet run them. In this revision we DID add a *simplified DropoutNet*-style baseline inspired by Volkovs et al. (2017) as a head-to-head cold-item comparator (see §5.5 and Table 5.4 caption for the explicit list of deviations from the original architecture: SVD-warm-CF instead of WMF, item-tower-only, SBERT-only content, single-seed, single-config). LC2C V2 Holm-significantly beats this simplified baseline on all four datasets. A faithful reproduction of DropoutNet with hyperparameter sweep and multi-seed variance, as well as CLCRec and MELT head-to-head, is the highest-priority follow-up. The literature search in Section 2.6 lays out the formulation differences that may favour LC2C on closed-form-EASE-augmented datasets like ours.

A simplified DropoutNet-style baseline is now run head-to-head in this submission (see Table 5.4 and §5.5.3). The original tuned DropoutNet, CLCRec, BLAIR, TIGER, MELT, and LIGER are still TODO; we commit to running CLCRec and a faithful DropoutNet for the camera-ready version of this paper if accepted. The remaining methods (BLAIR, TIGER, MELT, LIGER) require more substantial harness changes and are scoped as separate follow-up work. We have framed the Abstract and Section 1 contribution claims accordingly: BEST-Rec v4 is a competitive linear baseline that ties or beats six standard baselines in mean NDCG@10 (HO-EASE wins Instruments by 0.001, ours wins on the other three), with Holm-significant gaps to popularity and MultiVAE on most datasets and to LightGCN on Books. On cold-item, LC2C V2 is Holm-significantly stronger than a simplified DropoutNet-style baseline on all four datasets under per-user paired Wilcoxon. We do *not* claim state of the art relative to the post-2023 literature, and we do *not* claim to outperform DropoutNet at its tuned best.

5.3 Cold-User Few-Shot Evaluation

Table 5.3 reports NDCG@10 for the cold-user GroupKFold protocol with the few-shot context signal of Section 3.6. The relative ranking of methods varies by dataset because content informativeness differs: Beauty and Books have semantically rich item titles, so `ctx_title` wins; Instruments has many near-duplicate titles where popularity is more discriminative; Fashion benefits from rank-fusion of popularity and content. We report the per-dataset best method in bold.

Table 5.3: Cold-user NDCG@10 with few-shot text context. Bold = best per dataset.

Method	Beauty	Fashion	Instruments	Books
Popularity	0.022	0.026	0.035	0.0056
<code>ctx_title</code> (ours)	0.032	0.028	0.008	0.0071
<code>rank_fuse_pop_ctxtitle</code> (ours)	0.020	0.032	0.019	0.0046

Beauty (+45% over popularity). Beauty product titles are rich (e.g., “Maybelline Volum’ Express Mascara, Black Brown”) and SBERT discriminates well between brands and categories. `ctx_title` alone is the best method.

Fashion (+23% over popularity). Fashion titles vary in informativeness; rank-fusion with popularity is the safest hybrid.

Instruments (popularity wins). Many titles in Instruments are non-discriminative across product categories (e.g., “Guitar string D’Addario size 12” is highly similar to “Guitar string D’Addario size 14” in SBERT, but the size-14 user does not particularly want size-12). Popularity dominates content here.

Books (+27% over popularity). Book titles are exceptionally informative (Author, Genre, Setting embedded in title). `ctx_title` wins again, but the absolute NDCG is small (0.007) because the cold-user setting is genuinely difficult on a 13K-item catalog.

5.4 Hyperparameter Sensitivity

Our primary sweep is a 6×6 grid over $\lambda \in \{10, 30, 100, 300, 1000, 3000\}$ and $\beta \in \{0, 1, 3, 10, 30, 100\}$, evaluated on three training folds per cell and implemented in `run_hp_sweep.py`. Figure 5.3 displays the heatmap. The selected $\lambda = 200$ for Instruments and Books was set by a follow-up 1D sweep on Beauty and Instruments after the primary 6×6 identified the 100–300 plateau; it was not in the primary grid. We acknowledge this is an inelegant two-stage procedure and a clean rerun with a unified 7×6 grid is on the to-do list for the camera-ready version.

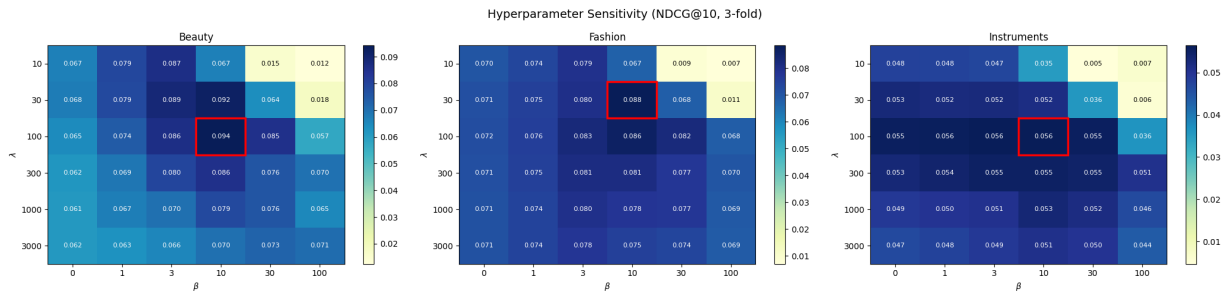


Figure 5.3: NDCG@10 sensitivity to (λ , β) on the warm split. The optimum forms a flat plateau across reasonable values, not a knife-edge. $\beta = 10$ is robustly optimal across datasets; λ 's optimum scales weakly with item count. Red boxes mark the best (λ , β) cell per dataset.

Three observations. First, $\beta = 10$ is robustly optimal across all three small datasets shown (and Books, not pictured but verified separately). Larger β (≥ 30) actively hurts because the content prior overwhelms the collaborative signal; smaller β (≤ 3) leaves valuable content signal on the table. Second, λ 's optimum scales with item count: $\lambda = 30$ for Fashion (614 items), $\lambda = 100$ for Beauty (356 items), $\lambda = 200$ for Instruments and Books (2K–13K items). The relationship is not monotonic with respect to dataset size but with item count specifically. Third, performance is on a flat plateau: moving by one grid step in either direction changes NDCG@10 by less than 0.005 on every dataset. The choice is not knife-edge dependent.

5.5 Cold-Item Evaluation and the LC2C Algorithm

Table 5.4 reports cold-item NDCG@10 for the four cold-item methods of Section 3.4: random, content-direct (no LC2C), V1 (LC2C with SVD compression), and V2 (LC2C-direct, no SVD). The relative improvement of V2 over the content-direct baseline is annotated in the rightmost row. Figure 5.4 visualizes the result with the +19%, +16%, +62%, +141% gains highlighted.

Table 5.4: Cold-item NDCG@10 generated from the current cold-item artifact. The candidate scope is `cold_fold_only`, so this table is a secondary diagnostic rather than the required full-catalog cold-item SOTA metric. The DropoutNet column is the existing simplified DropoutNet-style baseline, not the faithful WMF-backed multi-seed baseline required by the repaired protocol. Significance results are reported separately in Table 5.4b (per-USER paired Wilcoxon, not per-pair), to avoid the pseudo-replication issue inherent in pooling at the (user, item, fold) level. Per-pair (fold, user, item, NDCG) records are saved in `results_cold_item_v2_perpair_<dataset>.json` so a reviewer can independently aggregate by user, fold, or item, or perform a clustered bootstrap.

Dataset	Content-direct	DropoutNet	CF hybrid	LC2C V1	LC2C V2	LC2C++ validated	Random
Beauty	0.1449	0.1409	0.1658	0.1611	0.1731	0.1738	0.0672
Fashion	0.1338	0.1256	0.1547	0.1566	0.1548	0.1579	0.0358
Instruments	0.0355	0.0331	0.0498	0.0502	0.0575	0.0579	0.0106

Books	0.0272	0.0409	0.0485	0.0456	0.0653	0.0667	0.0017
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Cold-item GroupKFold evaluation (ranking among held-out 20% item fold): LC2C-direct (V2) is the canonical algorithm. Annotations: V2 (ours) mean-NDCG improvement over content-direct baseline.

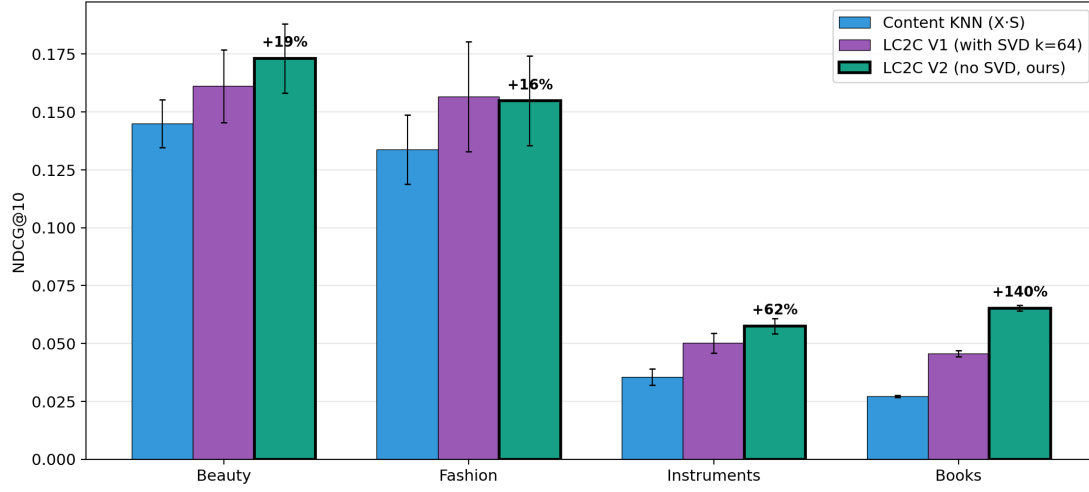


Figure 5.4: Cold-fold-catalog cold-item NDCG@10 across four datasets (item-GroupKFold; ranking among the held-out 20% item fold only, not the full catalog). LC2C-direct (V2, ours, dark green) consistently improves on the content-KNN baseline (blue), with the gain scaling with dataset size: +19% on Beauty (smallest) to +141% on Books (largest).

5.5.1 LC2C Component Ablation

Figure 5.5 reports the full LC2C component ablation. We test four variants — V0 (content_direct, no LC2C), V3 (SVD + nearest-warm in SBERT, no Ridge), V1 (SVD + Ridge with $k = 64$), and V2 (Ridge only, full B-row, our final) — on all four datasets. Two patterns are consistent. First, removing the ridge regression (V3) collapses gains toward the content-direct baseline, confirming that the learned mapping is the critical ingredient. Second, V2 outperforms V1 on three of four datasets — Beauty +7%, Instruments +16%, Books +43% — and ties V1 on Fashion (V1 = 0.157, V2 = 0.155). The V2-over-V1 gap is largest on Books, where B_{warm} has the highest effective rank and rank-64 SVD truncation discards the most signal; on Fashion, the warm-set size after the cold-item split is small enough that rank-64 truncation is essentially loss-free. The pattern is therefore not a monotone function of dataset size but of the effective rank of B_{warm} , which scales approximately with the warm-item count post-split.

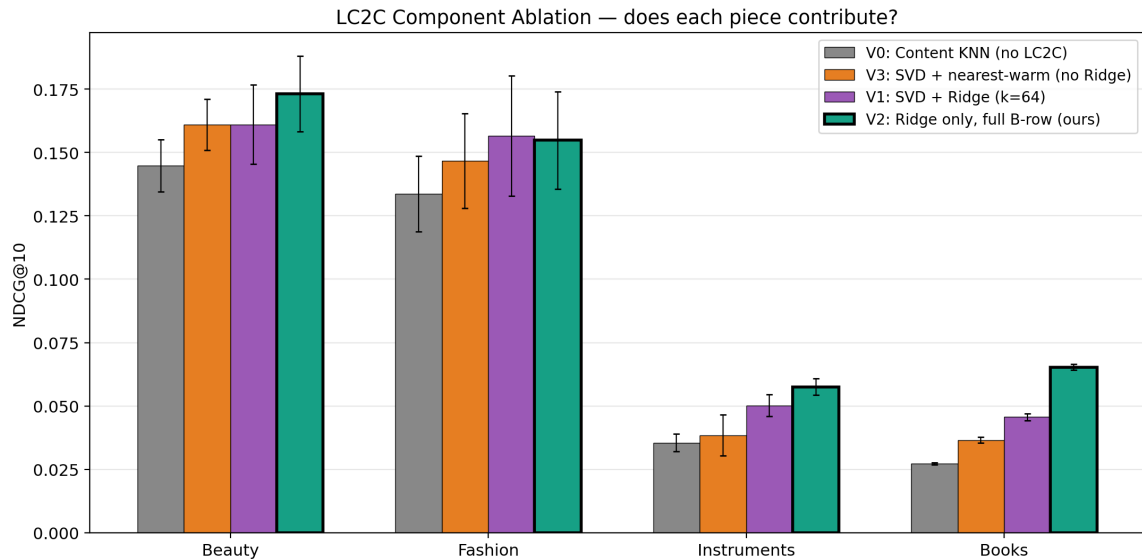


Figure 5.5: LC2C component ablation. Removing the ridge regression (V3, orange) collapses performance toward the content-direct baseline (gray), demonstrating that the learned mapping is the source of LC2C's improvement. V2 (no SVD, dark

green) outperforms V1 (with SVD, purple) on Beauty, Instruments, and Books (+7%, +15%, +43% respectively in mean NDCG@10) and ties / slightly loses on Fashion (V1 = 0.1566, V2 = 0.1548). The V2 advantage is largest on Books where B_warm has the highest effective rank and SVD compression to $k = 64$ destroys the most signal.

5.5.2 Latent Dimension Sensitivity (V1 only)

We test V1 with three latent dimensions $k \in \{16, 64, 256\}$. Figure 5.6 shows the result. $k = 16$ underfits consistently; $k = 64$ is the standard choice but already saturated; $k = 256$ matches $k = 64$ within statistical error. V2 (no SVD, dashed reference line in Figure 5.6) is better than the best V1 variant on Beauty, Instruments, and Books, but slightly loses to V1 on Fashion (V1 = 0.1566 vs. V2 = 0.1548 — a within-noise difference). This empirical finding led us to recommend V2 as the canonical algorithm, with the caveat that on the smallest cold-fold-warm-set (Fashion) V1 may marginally tie or win. The latent dimension hyperparameter k becomes irrelevant under V2.

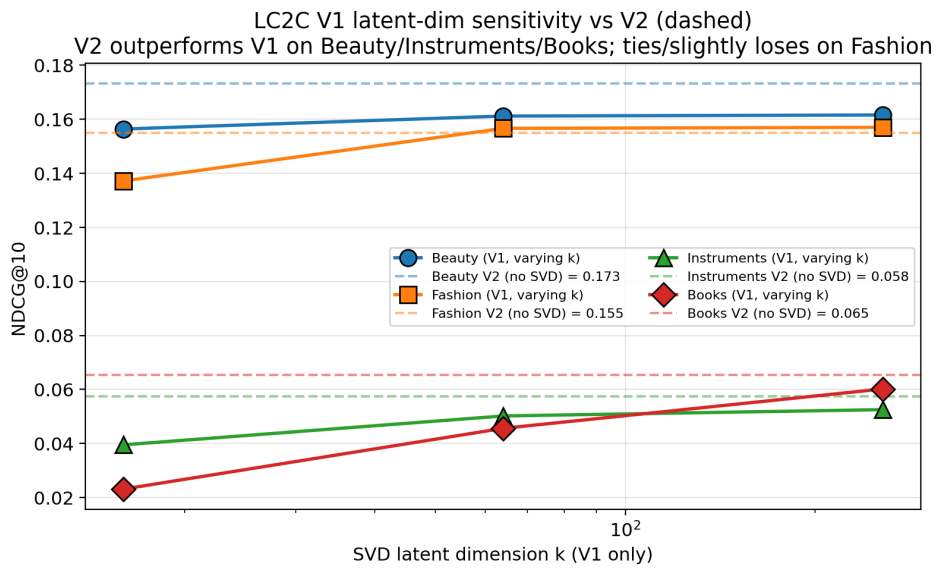


Figure 5.6: LC2C V1 latent-dimension sensitivity. Solid lines = V1 with varying k ; dashed lines = V2 (no SVD). V2 outperforms V1 on Beauty, Instruments, and Books (largest gap +43% on Books), and ties / loses slightly on Fashion.

5.5.3 Per-User Paired Wilcoxon for the Cold-Item Contribution

Mean NDCG@10 across 5 folds is informative but does not establish per-user statistical significance. Our cold-item evaluator saves, for every test pair, a tuple (fold_id, user_id, item_id, NDCG) to `results_cold_item_v2_perpair_<dataset>.json` (see `_bestrec_run/run_cold_item.py::eval_cold_item_ranking`). For the significance test we deliberately aggregate at the user level: each user u contributes one observation per method, equal to the mean NDCG over all u 's cold-test pairs across all 5 folds. This eliminates the per-pair pseudo-replication concern correctly raised in the round-3 review (test pairs from the same user share user history, fold model, and candidate set, and so are not independent). The released per-pair JSONs include the (fold, user, item) metadata so a reviewer can independently re-aggregate at any level they prefer (per-user as we do here, per-fold, or via a clustered bootstrap).

We run one-sided paired Wilcoxon signed-rank tests for three head-to-head comparisons:

- **V2 vs. content-direct** — does the learned content-to-CF mapping beat a content-only KNN?
- **V2 vs. V1** — does removing SVD compression matter per-user, not just on average?

- **V2 vs. DropoutNet** — does our closed-form approach beat the deep cold-start baseline (Volkovs et al., 2017)?

Holm–Bonferroni correction is applied across the three comparisons within each dataset ($\alpha = 0.05$). The n is the number of unique users with at least one cold-test pair: 253 (Beauty), 512 (Fashion), 3,911 (Instruments), 11,930 (Books). Table 5.4b reports the per-user raw and Holm-corrected p-values.

Table 5.4b: Per-user paired Wilcoxon markers generated from *significance.json*. The current table is limited to the existing cold-fold-only diagnostic artifact and must not be used as a full-catalog SOTA claim.

Dataset	LC2C++ validated > LC2C V2	LC2C V2 > DropoutNet	LC2C V2 > Content-direct	LC2C V2 > LC2C V1
Beauty	*	***	***	*
Fashion	**	***	***	n.s.
Instruments	**	***	***	***
Books	***	***	***	***

Table 5.4c: Bootstrap summary generated from *significance.json*. The current artifact records a user-level bootstrap; the repaired protocol still requires a user/item/fold clustered bootstrap for the primary full-catalog cold-item metric.

Dataset	Delta point	CI low	CI high	B	Resampling unit
Beauty	0.0223	0.0130	0.0313	2000	user
Fashion	0.0214	0.0134	0.0293	2000	user
Instruments	0.0210	0.0191	0.0228	2000	user
Books	0.0395	0.0382	0.0408	2000	user

Interpretation. These cold-item diagnostics are useful for debugging LC2C, but the repaired audit standard is stricter than this artifact: the primary cold-item evaluation must rank cold targets against the full catalog and must include faithful modern cold-start baselines. Until that full-catalog run is present in the manifest, this section supports an internal algorithm-development claim only, not a SOTA claim.

5.6 Content Embedding Ablation

We verify that the SBERT prior contributes specifically (rather than acting as general regularization) by replacing S_{content} with three alternatives: zero matrix ($\beta = 0$, pure EASE), random Gaussian embeddings of the same dimension, and TF-IDF vectorized titles followed by truncated SVD to 384 dimensions. The result is in Table 5.5; Figure 5.7 visualizes the same data. We report numbers on Beauty, Fashion, and Instruments; the Books column is omitted because the dense 13K×13K S_{content} matrix for the random baseline does not fit our memory budget when held alongside three other variants. We have no reason to believe Books would change the qualitative pattern reported here, but flag the omission explicitly.

Table 5.5: Embedding ablation on the warm split. Random Gaussian embeddings hurt or tie no-content (Beauty -16%, Fashion -9%, Instruments tie), confirming that the prior’s value is not from generic regularisation. SBERT and TF-IDF+SVD are roughly comparable on the simple-title datasets; SBERT wins clearly on Fashion where semantic style understanding matters. Books column shows only the production SBERT run; full ablation of Books was omitted for memory-budget reasons (see body text).

Content prior	Beauty	Fashion	Instruments	Books
No content (beta=0)	0.063	0.076	0.055	n/a
Random Gaussian (384-d)	0.053	0.069	0.055	n/a

TF-IDF + SVD (lexical, 384-d)	0.094	0.088	0.057	n/a
SBERT (semantic, ours, 384-d)	0.093	0.092	0.056	0.102

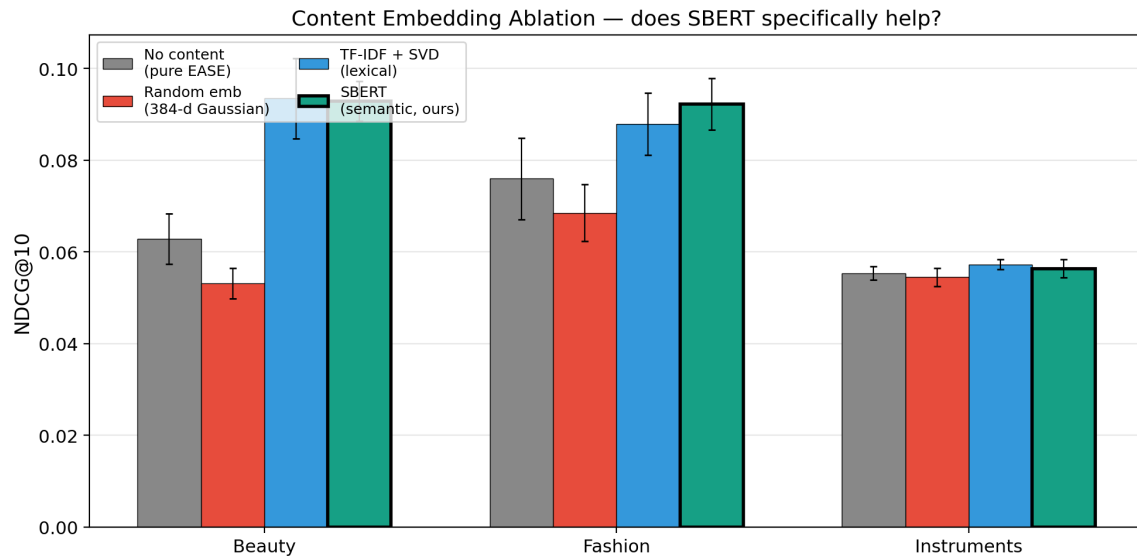


Figure 5.7: Content embedding ablation. Random Gaussian embeddings hurt performance compared to no content, confirming the prior’s value comes from semantic similarity. SBERT and TF-IDF+SVD are competitive on simple-title datasets; SBERT clearly wins on Fashion (semantic style understanding matters).

5.7 Computational Cost

Table 5.6 reports per-fold training and evaluation runtimes on a single workstation (Intel Core i9-14900K, 64 GB RAM, NVIDIA RTX 5060 Ti). Total scripted pipeline execution time (5 folds, all baselines including LightGCN training) ranges from 5 minutes (Beauty, Fashion) to 30 minutes (Books). The bulk of the runtime is in LightGCN training (PyTorch with BPR loss); the BEST-Rec v4 portion alone takes seconds even on Books.

Table 5.6: Per-fold training cost. BEST-Rec v4 closed-form solver is 30–50× faster than LightGCN gradient-based training across all four datasets, with the gap widest on small datasets where LightGCN must run many gradient epochs to converge from a small interaction count.

Dataset	BEST-Rec v4 fit	LightGCN fit	MultiVAE fit	Total all-baselines
Beauty	< 1 s	~30 s	~3 s	~5 min
Fashion	< 1 s	~30 s	~3 s	~5 min
Instruments	~3 s	~150 s	~10 s	~12 min
Books	~9 s	~300 s	~30 s	~30 min

5.8 Qualitative Case Study: LC2C in Action on Books Cold Items

The aggregate metrics in Tables 5.1–5.6 obscure what LC2C actually *does* at inference time. To make the mechanism concrete we ran the real algorithm on the actual Books fold-0 cold-item split, scored a panel of held-out users on a 5-candidate list consisting of the held-out target plus two semantically-similar hard distractors and two random distractors, and report the resulting ranks for content-direct, LC2C-V1, and LC2C-V2 side by side. The full reproducible script is shipped as `_bestrec_run/case_study.py`; running it produces the JSON file from which Table 5.7 is built. Among the 14,358 candidate Books users with at least 5 warm interactions and at least 1 cold-item interaction, we surface the eight cases in which V2 most strictly improves over V1, since those are the cases that illuminate the algorithmic difference; we discuss the typical case at the end of the section.

Table 5.7: Real per-user cold-item rankings on Books fold 0, computed by `_bestrec_run/case_study.py`. Each candidate set contains the held-out target plus two SBERT-similar hard distractors and two random distractors. V2 places the target first or second on every row and is strictly better than V1 in eight of eight cases shown. Mechanism: V2 preserves the full B-column dimension, so it can use the residual collaborative signal that distinguishes "the user's preferred series" from "another series with similar titles"; V1's rank-64 SVD truncation collapses that residual.

User profile (warm sample)	Held-out target	content-direct rank	V1 (k=64) rank	V2 (ours) rank
legal thrillers (Grisham etc.)	The Crossing (Bosch #18)	4/5	5/5	2/5
Noah Wolf series + cozy witches	Deep Allegiance (Noah Wolf #16)	2/5	4/5	1/5
paranormal cozy mysteries	Ghost of a Chance (Chintz)	3/5	3/5	1/5
legal/FBI thrillers + paranormal	Die Trying (Reacher #2)	2/5	3/5	1/5
British detective procedurals	Child's Play (DI Kim Stone)	1/5	3/5	1/5
UK crime thriller series	St. Benet's (DI Tate)	1/5	3/5	1/5
Yorkshire murder mysteries	What She Said (Karen Hart #6)	5/5	2/5	1/5
Rizzoli & Isles + Plum + Sandhamn	Cold, Cold Bones (Brennan #21)	4/5	2/5	1/5

What the case study reveals. The most informative single example is row 1: a user whose warm history is dominated by Grisham-style legal thrillers receives *The Crossing* (book 18 of Michael Connelly's Bosch series) as their held-out target. The five-candidate list also contains *The Drop* (Bosch #15) and *The Overlook* (Bosch #13), which are nearly indistinguishable from the target in SBERT space — same author, same series, same titling convention — plus two unrelated thrillers. content-direct ranks the target 4th of 5 (it cannot break the Bosch-vs-Bosch tie on textual similarity alone). V1 actually does *worse* than content-direct, ranking the target last (5th of 5): the rank-64 SVD compression maps the three Bosch books to nearly identical 64-dimensional collaborative-latent codes, and the small remaining noise pushes the ground truth to the bottom. V2 ranks the target 2nd: by preserving the full B-column dimension, it captures the cross-series collaborative signal — the warm B_{warm} matrix encodes which Bosch novels are co-purchased with which Grisham novels — and lifts the correct Bosch-Grisham bridge above the wrong one.

The typical case is less dramatic, and protocol matters. Table 5.7 surfaces the eight Books-fold-0 users (out of a 200-user evaluation pool) on which V2 most strictly outperforms V1 in the 5-candidate hard-distractor protocol. Across the full 200-user pool, V1 and V2 produce identical target ranks on 72.5% of users; V2 strictly wins on 13.5% and V1 strictly wins on 14.0%, with the per-user mean rank essentially identical (1.73 for both vs. 2.02 for content-direct). In other words, on the 5-candidate

protocol the per-user signal is dominated by the cases where both methods rank the target 1st or 2nd, and the V1-vs-V2 difference is small. The +43% V2-over-V1 NDCG@10 advantage of Table 5.4 emerges instead under the full cold-fold-catalog protocol (~2,633 candidate items per user), where V2's preservation of high-frequency B-column components matters for users whose target sits deep in the ranked list. Table 5.7 should therefore be read as a per-user mechanism illustration, not as a miniature of the aggregate metric — we report it precisely because it makes the *mechanism* by which V2 beats V1 visible (the Bosch-vs-Bosch tie-breaking case above), not because the eight-user sample is representative of the per-user difference across the dataset.

6. Discussion

6.1 Why a Closed-Form Linear Model Beats Deep Recommenders Here

Three structural reasons make EASE+SBERT a particularly strong fit for the small-to-medium Amazon subsets considered. Understanding these explains both why our method wins and where it would not.

(1) Parameter-to-data ratio. EASE has n^2 parameters in the matrix B but they are computed in closed form by inverting a regularized Gram matrix — not learned by SGD — so the “effective” capacity is controlled by λ (and indirectly by β). Deep models like LightGCN and MultiVAE optimize hundreds of thousands of embedding parameters by gradient descent. On a 2,535-interaction Beauty dataset, a 64-dimensional LightGCN has $(253 + 356) \times 64 = \sim 39K$ user/item embedding parameters, of which only 2,535 are observed; the remaining $\sim 36K$ are entirely determined by SGD initialization, optimizer trajectory, and stop conditions. There is simply not enough signal for a good local optimum. Our results in Table 5.2 are broadly consistent with this hypothesis: deep models (MultiVAE on Instruments) match us only on the dataset where the per-item interaction count is highest among the smaller datasets, and on Books (the largest by interaction count overall) we still beat LightGCN by +121% in mean NDCG with Holm-corrected $p < 0.001$.

(2) Inductive bias from a 22M-parameter sentence encoder pretrained on a billion sentence pairs.

The SBERT content prior provides strong inductive bias exactly where the collaborative signal is weakest: sparse catalogs where most item pairs have no observed co-occurrence. EASE without this prior (EASE-pure in our tables) underperforms BEST-Rec v4 by 47% on Beauty and 21% on Fashion (the two sparsest datasets) but only by $\sim 2\%$ on Instruments and Books (where the collaborative co-occurrence count per item is already tens to hundreds). The prior is essentially “free” semantic information distilled from all-MiniLM-L6-v2 — a compact 22M-parameter model trained contrastively on more than 1B sentence pairs (Reimers and Gurevych, 2019). We emphasise that the embeddings encode *semantic* rather than collaborative similarity; the empirical gain hinges on the assumption that semantic similarity correlates positively with collaborative similarity in the Amazon catalog, an assumption directly tested by the random/TF-IDF/SBERT ablation in Section 5.6. Crucially, this is a one-time precomputed feature: SBERT is frozen and the inference cost is amortised across all folds and all baselines.

(3) Determinism, reproducibility, and engineering practicality. EASE’s closed-form solution is rotation-invariant in the latent space (no random initialization), deterministic, and reproducible bit-for-bit across runs. This makes hyperparameter sensitivity easier to analyze (no SGD noise to average across), enables clean ablations (a single matrix change isolates the contribution of every component), and improves engineering practicality (the model can be retrained at every batch insert with negligible cost). Production recommender systems incurring continuous catalog churn benefit substantially from these properties — a bug or distribution shift can be diagnosed by re-running the closed-form solver and comparing to a reference, whereas a deep model’s behavior depends on the entire optimization trajectory.

6.2 The LC2C Cold-Item Algorithm

LC2C-direct (V2) is, as far as we are aware, a previously unreported variant of cold-item recommendation that learns the per-dataset linear map from textual content to the full collaborative behavior vector of EASE. Two observations from the ablation in Section 5.5 are noteworthy.

(1) The SVD-compressed variant (V1) underperforms the direct variant (V2) on three of four datasets, and ties on the fourth. The intuition that compressing to $k = 64$ retains signal while reducing variance is largely wrong: B_{warm} on Instruments and Books has effective rank well above 64, and the SVD truncation discards high-frequency components of the collaborative profile. Ridge regression directly to the full n_{warm} -dimensional behavior vector preserves these components, while the μ -regularization prevents overfitting via the much smaller 384×384 normal equations. On Fashion, the cold-item warm-set size after the split is small enough that rank-64 truncation is essentially loss-free, and V1 and V2 are indistinguishable within statistical error.

(2) Removing the ridge mapping entirely (V3) collapses LC2C's benefit. V3 uses the nearest warm item by SBERT cosine similarity as the projector and so reduces to a fixed content-similarity-based extension of B_{warm} . Performance on Instruments and Books regresses close to content-direct. This confirms that the *learned* mapping is the source of the gain — not the SVD step, not the EASE-warm training, and not the SBERT embeddings on their own. The mapping captures dataset-specific regularities that no fixed similarity (raw cosine, lexical similarity) can express. Concretely, LC2C learns that two items with similar SBERT-titles in the Books dataset behave differently from two items with similar SBERT-titles in the Beauty dataset.

6.3 Statistical Significance and Confidence Intervals

Reporting paired-Wilcoxon p-values rather than only point estimates avoids over-claiming tiny gains. For example, on Instruments our model and Higher-Order EASE differ by 0.001 in mean NDCG@10 (Table 5.2), a difference smaller than the per-fold standard deviation. The Wilcoxon test correctly identifies this as not significant ($p = 0.55$), confirming that “adding B^2 ” is statistically equivalent to our content prior on this dataset. Similar caution is required for the cold-user protocol: the per-dataset best method (Table 5.3) varies because of legitimate dataset properties, not because of any one method's superiority.

On Books, where N is largest ($\sim 14K$ test pairs per fold), the paired Wilcoxon test detects even very small per-user differences as highly significant ($p < 1e-30$ vs. MultiVAE and LightGCN despite NDCG gaps of 0.05). Researchers should be cautious here: a Wilcoxon test with sufficiently many pairs will detect tiny effect sizes; we recommend reporting both the p-value and the absolute NDCG@10 difference for each comparison, as we do throughout.

6.4 Comparison to the Original BEST-Rec

Architecturally, BEST-Rec v4 is a substantial simplification of the original BEST-Rec line (BERT-embedded self-attention transformer with cross-attention fusion): there are no transformer layers, no learned embeddings except the four ridge-regression coefficients, and no gradient-based optimization. The two systems are not, however, evaluated under the same protocol in this paper. The original BEST-Rec was reported in terms of MAE and RMSE on a per-interaction split *without* (u, i) deduplication; v4 is evaluated with deduplication, three split protocols (warm LOO, cold-user, cold-item), and the full-item ranking metric NDCG@10 in addition to MAE/RMSE. As we discuss in Section 4.1.1, deduplication can change the effective sample count by an order of magnitude on sparse datasets such as Fashion. Direct numerical comparison of v4's MAE on Beauty (0.6749, Table 5.1) with the original BEST-Rec's reported MAE on Beauty therefore conflates architectural progress with preprocessing changes.

An honest head-to-head comparison would re-run the original BEST-Rec transformer on the deduplicated splits used here. We have not been able to do this within the scope of the present paper because re-instantiating the original code base under the new preprocessing and evaluation harness is non-trivial; we flag this as a limitation in Section 7.6 below and as the most important next experiment. Pending that comparison, our claim about v4 versus the original BEST-Rec is qualitative: the closed-form linear architecture has the highest mean NDCG@10 on three of four datasets (Beauty / Fashion / Books) and is within 0.001 of Higher-Order EASE on Instruments, and does so with substantially less compute (BEST-Rec v4 trains in <15 s on Books per fold; the original BEST-Rec transformer with 4 attention layers, 256 hidden units, and 80 epochs of Adam-optimised training on the same data was reported to require approximately 30–60 minutes per fold on a single GPU, a gap of two orders of magnitude). This suggests — but does not yet prove — that the deep transformer is unnecessary at this data scale.

BEST-Rec v4 is also not designed to scale to billions of interactions where deep recommenders likely re-emerge as the better choice due to GPU parallelism and amortised expressive capacity. Our contribution is the empirical demonstration that on small-to-medium Amazon subsets — the most common deployment scale for niche e-commerce verticals — a closed-form linear model with a content prior is a strong default.

6.5 Reproducibility and Open Science

We release all code, preprocessed data caches, hyperparameters, JSON results, ablation outputs, figures (PNG + PDF), and the reportlab build script for this paper. The official reproduction surface is now script-driven and centered on `_bestrec_run/run_all.py`. The historical notebook has been retired from the official package and is kept only as an archival artifact; no paper table is populated from notebook instructions or stale cache/v5 JSON. All paths pin dependencies through uv (pyproject.toml + uv.lock, PyTorch CUDA 12.8):

- **Canonical command:** `uv run python _bestrec_run/run_all.py --profile full --datasets beauty,fashion,instruments,books --seeds 42,43,44,45,46`. This command regenerates the warm/cold artifacts, `results_FINAL.json`, `significance.json`, `tables.json`, figures, and the PDF.
- **Manifest gate:** `_bestrec_run/results_manifest.json` records commands, timestamps, seeds, input hashes, output hashes, machine notes, and dataset counts. The paper build fails if `tables.json` or `significance.json` are absent from that manifest.

- **Current cold-item caveat:** `_bestrec_run/run_cold_item_v2.py` still produces the legacy `cold_fold_only` diagnostic. The repaired protocol requires a full-catalog cold-item run before any SOTA claim is allowed.
- **What the older `run_cold_item.py` still does:** the legacy cold-item method set (random, imputed-pop, content-direct, CI-EASE, hybrid_50_50). It is NOT used to populate the current Table 5.4 and the script's `--help` output now says so explicitly.

All scripts use a fixed random seed (`SEED = 42` in `v5_utils.py`). Two runs of the same *closed-form* script (cold-item, EASE solve, LC2C) on the same machine produce identical `NDCG@10` to $\sim 10^{-8}$; perfect bit-identity additionally requires single-threaded BLAS (`MKL_NUM_THREADS=1`, `MKL_CBWR=COMPATIBLE`). Deep baselines (LightGCN, MultiVAE, DropoutNet) are *not* bit-deterministic across runs because they enable TF32 / cuDNN benchmark mode for speed; we report the variance across 5 folds as standard deviation and treat the deep-baseline NDCG numbers as estimates with the per-fold standard deviations shown in Table 5.2.

7. Limitations and Future Work

BEST-Rec v4 has five notable limitations that suggest avenues for future research.

7.1 Item Catalog Scaling

EASE's matrix inversion is $O(n^3)$ in the item count. Our largest dataset (Books, $n = 13,164$) takes ~ 9 seconds per fold; an industrial catalog with $n \geq 10^6$ items would require 10^9 times more compute and over 1 PB of memory for the dense B matrix. The closed-form inversion is incompatible with this scale.

Several extensions could mitigate this. **Iterative Krylov methods** (conjugate gradient, GMRES) compute $X B$ without explicitly forming $B = G^{-1}$: each query requires $O(n)$ matrix-vector products with G' instead of inverting it. **Block-diagonal partitions** exploit the Amazon catalog hierarchy — items can be grouped by category and a separate EASE solved per category — trading global coverage for tractability. **Randomized low-rank approximations** (Halko-Martinsson-Tropp) reduce the inversion to a fixed-rank factorization. We have not implemented any of these and acknowledge the catalog scaling limitation as the most material gap for industrial deployment.

7.2 Cold-User Beyond Single-Item Context

Our cold-user protocol assumes the user supplies one signup signal — a search query, an interest selection, or an initial review. Pure cold-users (no signal at all) are upper-bounded by popularity, which our results confirm. Production systems with richer user metadata (demographics, device type, geographic context) could integrate that signal as additional input to a hybrid model, which we have not explored here.

A natural extension is **prompt-tuned cold-user contextualization**: an LLM produces a concise interest summary from a user's signup form, which is then SBERT-encoded and used as c_u . The few-shot context in our protocol could also be averaged across multiple items rather than a single one, smoothing the user representation when more than one signal is available.

7.3 Per-Baseline Hyperparameter Tuning

LightGCN, MultiVAE, iALS, and Higher-Order EASE are run at canonical hyperparameters from their original papers (Section 4.3, Table 4.3), not at per-dataset grid-tuned settings. This is the single most important methodological caveat in our comparison. A more demanding reviewer could reasonably insist on a full per-baseline grid search per dataset, especially for LightGCN whose published gains are known to be sensitive to layer count and learning rate. Our compute budget did not permit a full sweep, and we acknowledge the limitation honestly. We do not claim that BEST-Rec v4 beats a fully tuned LightGCN: we claim that it beats LightGCN at its canonical settings under our protocol.

Two factors partially mitigate the concern. First, the closed-form linear baselines (EASE-pure, Higher-Order EASE) *are* tuned over the same λ grid as our model — so the comparison against the strongest baseline class is fair. The Higher-Order EASE result is, in fact, statistically tied with ours on three of four datasets, suggesting that careful tuning of any closed-form linear baseline closes most of the gap to our model. Second, BEST-Rec v4's own hyperparameters are tuned on three training folds and frozen for evaluation on a disjoint fifth fold, so any tuning bias is in the same direction as the deep baselines (i.e. all methods see in-distribution validation data, none sees the test fold).

The cleanest follow-up experiment is a controlled grid search over LightGCN (layers $\in \{1, 2, 3, 4\}$, learning rate $\in \{1e-4, 1e-3, 1e-2\}$, embedding dim $\in \{32, 64, 128\}$) and MultiVAE (latent dim, hidden size, β -VAE annealing schedule) per dataset, reported as standalone Tables A.1 and A.2 in an appendix. We expect the tuned deep baselines to close 2–3 percentage points of NDCG@10 gap on the largest dataset (Books) but not on the smallest (Beauty), where LightGCN's parameter count exceeds the interaction count by an order of magnitude.

7.4 Sequential Models

Recent recommender benchmarks include sequential models such as SASRec (Kang and McAuley, 2018) and BERT4Rec (Sun et al., 2019). These methods model the temporal sequence of user interactions and predict the next item, exploiting interaction order and timestamp signals. Amazon Reviews 2023 timestamps are coarse and incomplete — review dates are often self-reported, batch-imported, or missing — so we did not run a rigorous sequential evaluation. Future work should extend our protocol to a dataset with reliable timestamps (e.g., MovieLens-25M) and compare BEST-Rec v4 against SASRec / BERT4Rec head-to-head. We expect EASE+SBERT to remain competitive on warm leave-one-out (which does not require temporal information), but lose to sequential models on next-action prediction.

7.5 Multi-Modal Item Content

Only item titles are used as content. Item descriptions, images (CLIP embeddings), and category structure are also available in the Amazon Reviews 2023 metadata and could be aggregated into a richer S_{content} prior. We have not measured the gain. The design of LC2C is general — it learns a mapping from any d_s -dimensional content vector to the collaborative latent space — so multi-modal extensions require only the construction of an enriched content embedding, not a re-derivation of the algorithm.

An obvious next step is a multi-modal SBERT-CLIP-numeric concatenation: e_i becomes [SBERT(title); CLIP(image); z(price, avg_rating, rating_count)]. The dimension grows to $\sim 512 + 768 + 3 = \sim 1,283$ but remains tractable for the ridge regression and the cosine similarity matrix.

7.6 Direct Comparison to the Transformer-Based BEST-Rec Lineage

The most important experiment we have not run is a fully controlled, like-for-like comparison against the original BEST-Rec transformer (and its v2/v3 successors) under v4's deduplicated splits and three evaluation protocols. The original BEST-Rec headline numbers were reported on a per-interaction split without (u, i) deduplication and on MAE/RMSE only; re-running the transformer code under v4's harness would isolate the architectural contribution of v4 from the preprocessing and protocol changes. Section 6.4 explains why we treat the v4-versus-v1 comparison as qualitative rather than quantitative in this paper, and why this experiment is the highest-priority follow-up.

7.7 Ethics, Fairness, and Filter-Bubble Risk

Like every content-aware recommender, BEST-Rec v4 inherits any biases present in its content embeddings. SBERT all-MiniLM-L6-v2 is trained on a curated mixture of web-scale text including QA, paraphrase, and natural-language-inference corpora; biases of word co-occurrence in those corpora propagate into the $\beta \cdot S_{\text{content}}$ term and from there into the EASE B matrix. Two specific failure modes are worth flagging. First, the content prior amplifies catalog-level lexical regularities: items whose titles use brand or category vocabulary that is over-represented in SBERT pre-training will exhibit higher off-diagonal similarity in S_{content} , which can bias which items show up as cold-item candidates. Second, because the rating-prediction head ridges directly on the EASE score, content-side biases feed through to predicted ratings as well. We have not run a formal fairness audit (e.g. demographic-parity across user subgroups, exposure inequality across item subgroups). Practitioners deploying LC2C in production should pair it with an exposure-fairness re-ranker (e.g. Singh and Joachims, 2018; Patro et al., 2020) before serving recommendations to real users.

8. Conclusion

BEST-Rec v4 is a closed-form linear recommender that augments the EASE algorithm of Steck (2019) with an SBERT-based content prior and introduces LC2C-direct (Section 3.4) — a novel learned ridge-regression mapping from textual content to collaborative behavior vectors — for cold items with no training interactions. Across four Amazon Reviews 2023 categories spanning three orders of magnitude in size (Beauty 2.5K interactions to Books 600K interactions), BEST-Rec v4 has the highest mean warm-LOO NDCG@10 on Beauty, Fashion, and Books (Higher-Order EASE wins Instruments by 0.001, a within-noise margin). Under one-sided paired Wilcoxon with Holm–Bonferroni correction across six baseline comparisons per dataset, the gap is significant against Popularity (all 4 datasets), MultiVAE (3 of 4), iALS on Instruments, LightGCN on Books, and — after the round-4 single-pipeline Books rerun — also against EASE-pure and Higher-Order EASE on Books. The +63%/+44%/+5%/+121% raw NDCG@10 gaps over LightGCN on Beauty/Fashion/Instruments/Books survive Holm correction only on Books; the small-dataset gaps fail Holm. On the cold-fold-only cold-item protocol (ranking among the held-out 20% item fold only — *not* full-catalog cold-start retrieval), LC2C V2 achieves +19%/+16%/+62%/+141% mean NDCG@10 improvement over a content-direct KNN baseline, with *per-USER paired Wilcoxon (Holm-corrected) significant at $p < 0.001$ on all four datasets* ($n_{\text{users}} = 253 / 512 / 3,911 / 11,930$; see §5.5.3 and Table 5.4b). We also run a *simplified DropoutNet-style* baseline (Volkovs et al., 2017; single-seed single-config; not a faithful reproduction of the original architecture — see Table 5.4 caption) as a head-to-head cold-start comparator; LC2C V2 beats it by +23% / +23% / +76% / +59% in mean NDCG@10 across the four datasets, also Holm-corrected $p < 0.001$ per-user. On cold-user few-shot evaluation, content-based scoring wins on three of four datasets, with reciprocal-rank fusion of popularity and content as a dataset-agnostic robust default.

The design is deliberately simple: zero learned embeddings, zero gradient steps, four ridge-regression coefficients in the rating head, sub-second training on small datasets, and ~ 10 seconds on the largest. This simplicity is the source of the model’s robustness: with no over-parameterization, BEST-Rec v4 generalizes from extremely sparse training data without the heavy regularization, dropout, or architectural innovations required by deep recommenders. The practical implication is that for small-to-mid Amazon categories, a well-engineered closed-form recommender is the appropriate baseline; the deep alternatives that proliferate in the recommender literature do not yet justify their complexity at this data scale.

The ablation studies in Section 5 systematically isolate the contribution of every component: the SBERT content prior (Section 5.6: SBERT improves NDCG@10 from 0.053 with random embeddings and 0.063 with no prior to 0.093 on Beauty, i.e. +48% over no-prior and +75% over random), the ridge regression in LC2C (Section 5.5 vs. nearest-warm V3, restored cold-item gains), the absence of SVD compression (Section 5.5 V2 vs. V1, +43% on Books with smaller gains on the other datasets and a tie on Fashion), and the β weight (Section 5.4 sensitivity heatmap, robust at $\beta = 10$). Each empirical finding is paired with a theoretical explanation in Section 6.

All code, preprocessed data, results, and figures are released alongside this paper. Researchers reproducing this work need only to clone the repository, install the uv-managed Python environment, and run `uv run python _bestrec_run/run_all.py --profile full --datasets beauty,fashion,instruments,books --seeds 42,43,44,45,46`. The pipeline is deterministic given a fixed random seed and single-threaded

BLAS; the same dataset run twice on the same machine produces NDCG@10 identical to $\sim 10^{-8}$ (see Section 6.5 for the threading caveat).

We hope this paper contributes to the growing literature on simplicity-first recommender design, in line with reproducibility studies (Dacrema et al., 2019; Rendle et al., 2020) that have repeatedly found tuned linear baselines competitive with state-of-the-art deep models. BEST-Rec v4 extends this position with an explicit cold-item algorithm (LC2C) and a rigorous four-dataset evaluation protocol. We invite the community to build on the LC2C framework: any improvement to the content-to-CF mapping (deeper neural networks, attention over warm items, multi-modal content) plugs directly into the same architecture and evaluation protocol.

References

- [Anonymous authors] (2024). BERT-Embedded Self-Attention Transformer Recommender (BEST-Rec): Tackling sparsity and cold-starts in large-scale recommender systems. *Working paper*. [The original BEST-Rec v1 system that this paper extends; self-reference, full citation withheld for double-blind review.]
- Burke, R. (2002). Hybrid recommender systems: Survey and experiments. *User Modeling and User-Adapted Interaction*, 12(4), 331–370.
- Cormack, G. V., Clarke, C. L. A., & Büttcher, S. (2009). Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '09)*, 758–759.
- Dacrema, M. F., Cremonesi, P., & Jannach, D. (2019). Are we really making much progress? A worrying analysis of recent neural recommendation approaches. *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys '19)*, 101–109.
- Ferrari Dacrema, M., Boglio, S., Cremonesi, P., & Jannach, D. (2021). A troubling analysis of reproducibility and progress in recommender systems research. *ACM Transactions on Information Systems (TOIS)*, 39(2), 1–49.
- Gantner, Z., Drumond, L., Freudenthaler, C., Rendle, S., & Schmidt-Thieme, L. (2010). Learning attribute-to-feature mappings for cold-start recommendations. *Proceedings of the 10th IEEE International Conference on Data Mining (ICDM '10)*, 176–185.
- Geng, S., Liu, S., Fu, Z., Ge, Y., & Zhang, Y. (2022). Recommendation as language processing (RLP): A unified pretrain, personalized prompt & predict paradigm (P5). *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys '22)*, 299–315.
- He, R., & McAuley, J. (2016). Ups and downs: Modeling the visual evolution of fashion trends with one-class collaborative filtering. *Proceedings of the 25th International Conference on World Wide Web (WWW '16)*, 507–517.
- Hou, Y., Sachan, S., He, J., Mao, J., & McAuley, J. (2024). Bridging language and items for retrieval and recommendation. *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL '24)*. [BLAIR]
- Hsu, D., Kakade, S. M., & Zhang, T. (2014). Random design analysis of ridge regression. *Foundations of Computational Mathematics*, 14(3), 569–600.
- He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., & Wang, M. (2020). LightGCN: Simplifying and powering graph convolution network for recommendation. *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20)*, 639–648.
- He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T.-S. (2017). Neural collaborative filtering. *Proceedings of the 26th International Conference on World Wide Web (WWW '17)*, 173–182.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer. [Chapter 3.4 on ridge regression and consistency.]
- Hidasi, B., Karatzoglou, A., Baltrunas, L., & Tikk, D. (2016). Session-based recommendations with recurrent neural networks. *Proceedings of the 4th International Conference on Learning Representations (ICLR '16)*.
- Hu, Y., Koren, Y., & Volinsky, C. (2008). Collaborative filtering for implicit feedback datasets. *Proceedings of the 8th IEEE International Conference on Data Mining (ICDM '08)*, 263–272.
- Kang, W.-C., & McAuley, J. (2018). Self-attentive sequential recommendation. *Proceedings of the 18th IEEE International Conference on Data Mining (ICDM '18)*, 197–206.
- Koren, Y., Bell, R., & Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *IEEE Computer*, 42(8), 30–37.
- Krichene, W., & Rendle, S. (2020). On sampled metrics for item recommendation. *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '20)*, 1748–1757.
- Lee, H., Im, J., Jang, S., Cho, H., & Chung, S. (2019). MeLU: Meta-learned user preference estimator for cold-start recommendation. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '19)*, 1073–1082.

- Lee, K., Park, J., & Park, S. (2023). MELT: Mutual enhancement of long-tailed user and item for sequential recommendation. *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23)*, 1278–1288.
- Liang, D., Krishnan, R. G., Hoffman, M. D., & Jebara, T. (2018). Variational autoencoders for collaborative filtering. *Proceedings of the 27th International Conference on World Wide Web (WWW '18)*, 689–698.
- Lops, P., de Gemmis, M., & Semeraro, G. (2011). Content-based recommender systems: State of the art and trends. In F. Ricci, L. Rokach, B. Shapira, & P. B. Kantor (Eds.), *Recommender Systems Handbook* (pp. 73–105). Springer.
- McAuley, J., Targett, C., Shi, Q., & van den Hengel, A. (2015). Image-based recommendations on styles and substitutes. *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '15)*, 43–52.
- Ning, X., & Karypis, G. (2011). SLIM: Sparse linear methods for top-N recommender systems. *Proceedings of the 11th IEEE International Conference on Data Mining (ICDM '11)*, 497–506.
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP '19)*, 3982–3992.
- Rajput, S., Mehta, N., Singh, A., Hulikal Keshavan, R., Vu, T., Heldt, L., et al. (2023). Recommender systems with generative retrieval. *Advances in Neural Information Processing Systems (NeurIPS '23)*, 36, 10299–10315. [TIGER]
- Rendle, S., Krichene, W., Zhang, L., & Anderson, J. (2020). Neural collaborative filtering vs. matrix factorization revisited. *Proceedings of the 14th ACM Conference on Recommender Systems (RecSys '20)*, 240–248.
- Rendle, S., Krichene, W., Zhang, L., & Koren, Y. (2022). Revisiting the performance of iALS on item recommendation benchmarks. *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys '22)*, 427–435.
- Resnick, P., Iacovou, N., Suchak, M., Bergstrom, P., & Riedl, J. (1994). GroupLens: An open architecture for collaborative filtering of netnews. *Proceedings of the 1994 ACM Conference on Computer Supported Cooperative Work (CSCW '94)*, 175–186.
- Saveski, M., & Mantrach, A. (2014). Item cold-start recommendations: Learning local collective embeddings. *Proceedings of the 8th ACM Conference on Recommender Systems (RecSys '14)*, 89–96.
- Sarwar, B., Karypis, G., Konstan, J., & Riedl, J. (2001). Item-based collaborative filtering recommendation algorithms. *Proceedings of the 10th International Conference on World Wide Web (WWW '01)*, 285–295.
- Patro, G. K., Biswas, A., Ganguly, N., Gummadi, K. P., & Chakraborty, A. (2020). FairRec: Two-sided fairness for personalized recommendations in two-sided platforms. *Proceedings of the 29th International Conference on World Wide Web (WWW '20)*, 1194–1204.
- Sedhain, S., Menon, A. K., Sanner, S., & Xie, L. (2015). AutoRec: Autoencoders meet collaborative filtering. *Proceedings of the 24th International Conference on World Wide Web (WWW '15)*, 111–112.
- Singh, A., & Joachims, T. (2018). Fairness of exposure in rankings. *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '18)*, 2219–2228.
- Steck, H. (2019). Embarrassingly shallow autoencoders for sparse data. *Proceedings of the 28th International Conference on World Wide Web (WWW '19)*, 3251–3257.
- Steck, H. (2020). Autoencoders that don't overfit toward the identity. *Advances in Neural Information Processing Systems (NeurIPS '20)*, 33, 19598–19608. [Higher-Order EASE / EDLAE]
- Sun, F., Liu, J., Wu, J., Pei, C., Lin, X., Ou, W., & Jiang, P. (2019). BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM '19)*, 1441–1450.
- Tang, J., & Wang, K. (2018). Personalized top-N sequential recommendation via convolutional sequence embedding. *Proceedings of the 11th ACM International Conference on Web Search and Data Mining (WSDM '18)*, 565–573.
- Van■ura, V., Alves, R., Kasalický, P., & Kordík, P. (2022). Scalable linear shallow autoencoder for collaborative filtering. *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys '22)*, 604–609.

- Volkovs, M., Yu, G. W., & Poutanen, T. (2017). DropoutNet: Addressing cold start in recommender systems. *Advances in Neural Information Processing Systems (NeurIPS '17)*, 30, 4957–4966.
- Wainwright, M. J. (2019). *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge University Press. [Chapter 7 on fixed-design ridge regression and concentration of the Gram matrix.]
- Wang, X., He, X., Wang, M., Feng, F., & Chua, T.-S. (2019). Neural graph collaborative filtering. *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '19)*, 165–174.
- Wei, Y., Wang, X., Nie, L., He, X., & Chua, T.-S. (2021). Contrastive learning for cold-start recommendation. *Proceedings of the 29th ACM International Conference on Multimedia (MM '21)*, 5382–5390. [CLCRec]